



INTRODUCTION TO PETROLEUM GEOLOGY FOR DRILLERS

Dr. Sena W. Reksalegora

Independent Consultant

Visiting Lecturer, Geology Department UNPAD

Sena.Reksalegora@gmail.com

+62 811 8000 467



This short course's primary goal is to give participants a basic knowledge of petroleum geology, which they can then integrate with engineering data to apply in the drilling environment as efficiently and effectively as possible.

Reference:

Magoon L.B. dan Dow W.G. (ed.) (1994) The Petroleum System – From Source to Trap. AAPG Memoir 60, AAPG, Tulsa.



About the Instructor

KEY COMPETENCIES:

Well seasoned petroleum geoscientist with 35+ years of productive industry experience mostly in major E&P companies in country and overseas (ARCO, UNOCAL, CHEVRON, APEC, HUSKY). Has held various technical positions such as Chief Geologist, Senior Advisor Geologist, Team Leader, Company Deputy Chief Rep.

WORKING EXPERIENCE:

- 1987 **ARCO INDONESIA**, Jakarta-Indonesia: Sr. Geologist/Team Leader – Exploration/Development
ARCO ALASKA – KUPARUK RIVER UNIT Anchorage - Alaska: Geologist
- 1999 **UNOCAL INDONESIA**, Balikpapan-Indonesia: Chief Geologist
- 2002 **UNOCAL DEEPWATER INTERNATIONAL – GULF OF MEXICO**, Sugarland-Texas: Consulting Geologist
- 2005 **CHEVRON DEEPWATER INDONESIA**, Houston-Texas: Sr. Advisor Geologist Deep Water Exploration/Development
- 2009 **APEC (Asia Pacific Exploration Consolidated)**, Jakarta-Indonesia: Deputy Chief Rep. Exploration and New Venture
- 2013 **HUSKY ENERGY**, Jakarta-Indonesia: Chief Geologist/Deputy Chief Rep. Exploration and New Ventures
- **2019-Present INDEPENDENT CONSULTANT** (Petroleum Prospectivity: East Kalimantan, Central Sumatera Basin, South Sumatera, SE Sulawesi, Buton, Papua, Tanimbar. Petroleum Geomechanics: East Java Basin, Kutai Basin)
VISITING LECTURER FTG UNPAD (Petroleum Geology, Petroleum Exploration, Petroleum Geomechanics, Basin Analysis)

EDUCATION:

- Undergraduate : Geological Science, University of Nevada Reno - USA
- Graduate/Post Graduate : Petroleum Geology/Overpressure and Tectonics, ITB

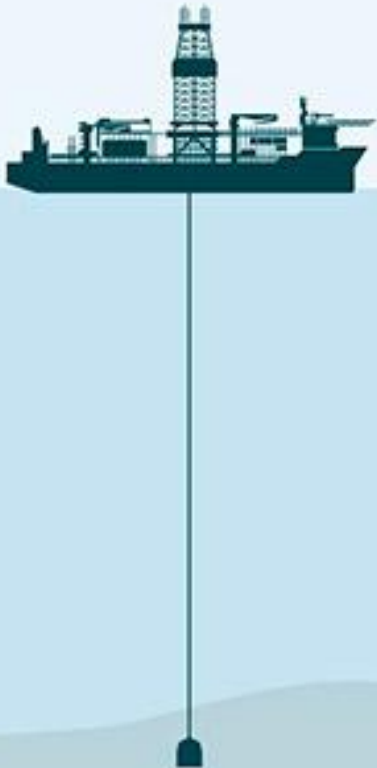
Agenda

- Introduction
- Petroleum System
 - Source Rocks and Hydrocarbon Migration
 - Reservoir
 - Trap and Seal
- Pore Pressure and In-situ Stress
- Prospect and Lead
- Subsurface Uncertainty and Drilling Risk

Types of Drilling Rigs

Drillship

Operates at water depths up to 12,000ft.



Semi-submersible

Operates at water depths up to 10,000ft.



Jack-Up Rig

Operates at water depths up to 500ft.



Drilling Barge

Operates in shallow waters.



Landrig



PETROMINER
PETROLEUM • MINING • ENERGY

Headlines Petroleum Upstream

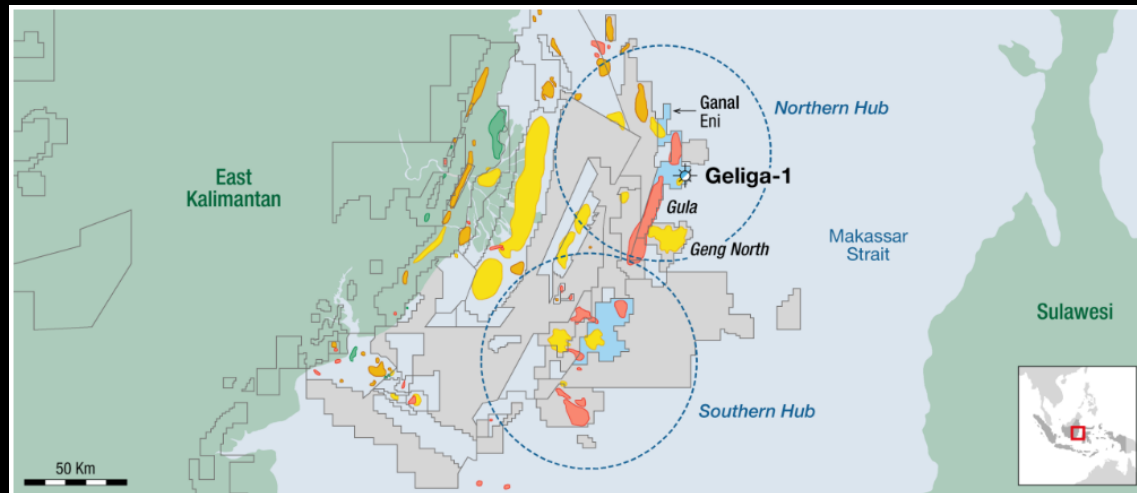
Berita Baik dari Sumur Eksplorasi Dara Jingga di CepuPosted by **Prismono** – 5 Mei 2026

Kegiatan uji produksi di sumur eksplorasi Dara Jingga (DRJ)-001 di Blora, Jawa Tengah.

Jakarta, Petrominer – Kegiatan pengeboran sumur eksplorasi Dara Jingga (DRJ)-001 di Blora, Jawa Tengah, diyakini berhasil menemukan migas. Berdasarkan uji produksi awal, ada aliran gas dengan tekanan cukup besar dan kondensat beberapa barel.

"Alhamdulillah, pengeboran eksplorasi yang dilakukan oleh Pertamina EP Zona-11, PHE Regional 4 berhasil menemukan migas dan sedang melaksanakan uji test pada sumur eksplorasi Dara Jingga-01 (DRJ-01)," tulis Kepala SKK Migas, Djoko Siswanto, dalam laporannya kepada Menteri ESDM yang diperoleh **PETROMINER**, Selasa (5/5).

Menurut Djoko, pengetesan dilakukan dengan DST#2 dan gas terlihat *flowing*, sekitar 11 mmcsfd dengan tekanan cukup besar. Status saat ini masih *clean up*, dan masih evaluasi besarnya kondensat.



Asia Oil & Gas

Eni's success in the Kutei Basin continues

<https://geoexpro.com/enis-success-in-the-kutei-basin-continues/>

Geliga-1 boosts Kutei Basin potential



By Jonathan Craig, NVentures

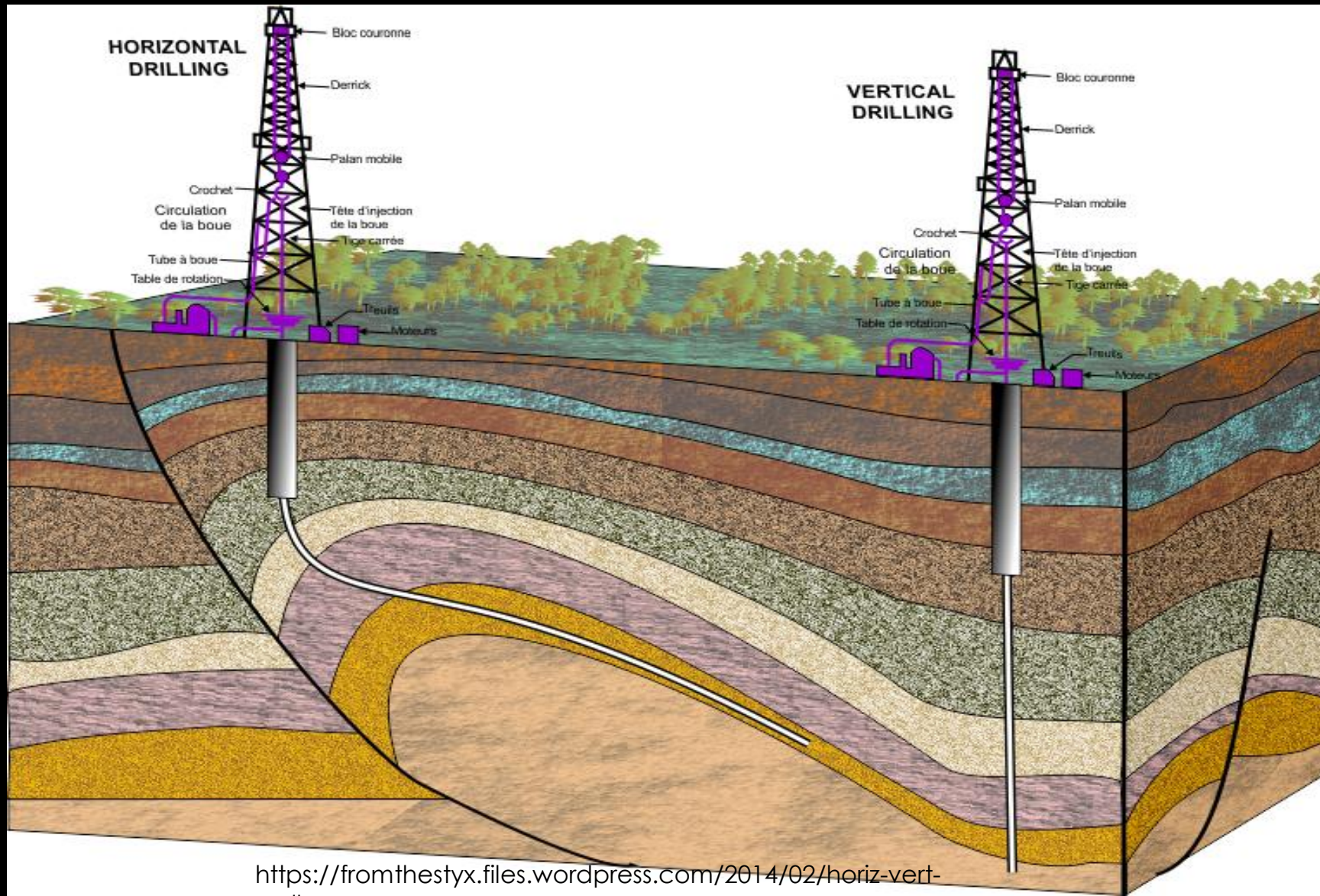
Published Date: April 28, 2026

Eni announced a major gas discovery at the **Geliga-1** wildcat in the Ganai block, offshore Indonesia's Kutei Basin in April 2026. The well, approximately 70km offshore, is estimated to contain around 5 Tcf of gas and 300 million barrels of condensate in place.

Drilled to ~5,100 m in 2,000m of water, Geliga-1 encountered a significant gas column in high-quality Miocene reservoirs, with drill stem test planned to assess productivity. The discovery follows recent successes including Geng North in 2023 and Konta-1 in 2025, reinforcing the basin's scale and potential.

Drilling Targets

- Hydrocarbon accumulation





CONVENTIONAL AND UNCONVENTIONAL HYDROCARBON

Definition

The difference between conventional and unconventional is relatively straightforward, and has to do mostly with the ease the fuels can be extracted with.

- **CONVENTIONAL** oil or gas comes from formations that are "normal" or straightforward to extract product from.
- **UNCONVENTIONAL** oil or gas resources are much more difficult to extract. Some of these resources are trapped in reservoirs with poor permeability and porosity, meaning that it is extremely difficult or impossible for oil or natural gas to flow through the pores and into a standard well. To be able to produce from these difficult reservoirs, specialized techniques and tools are used.



SEDIMENTARY BASIN DEFINITION

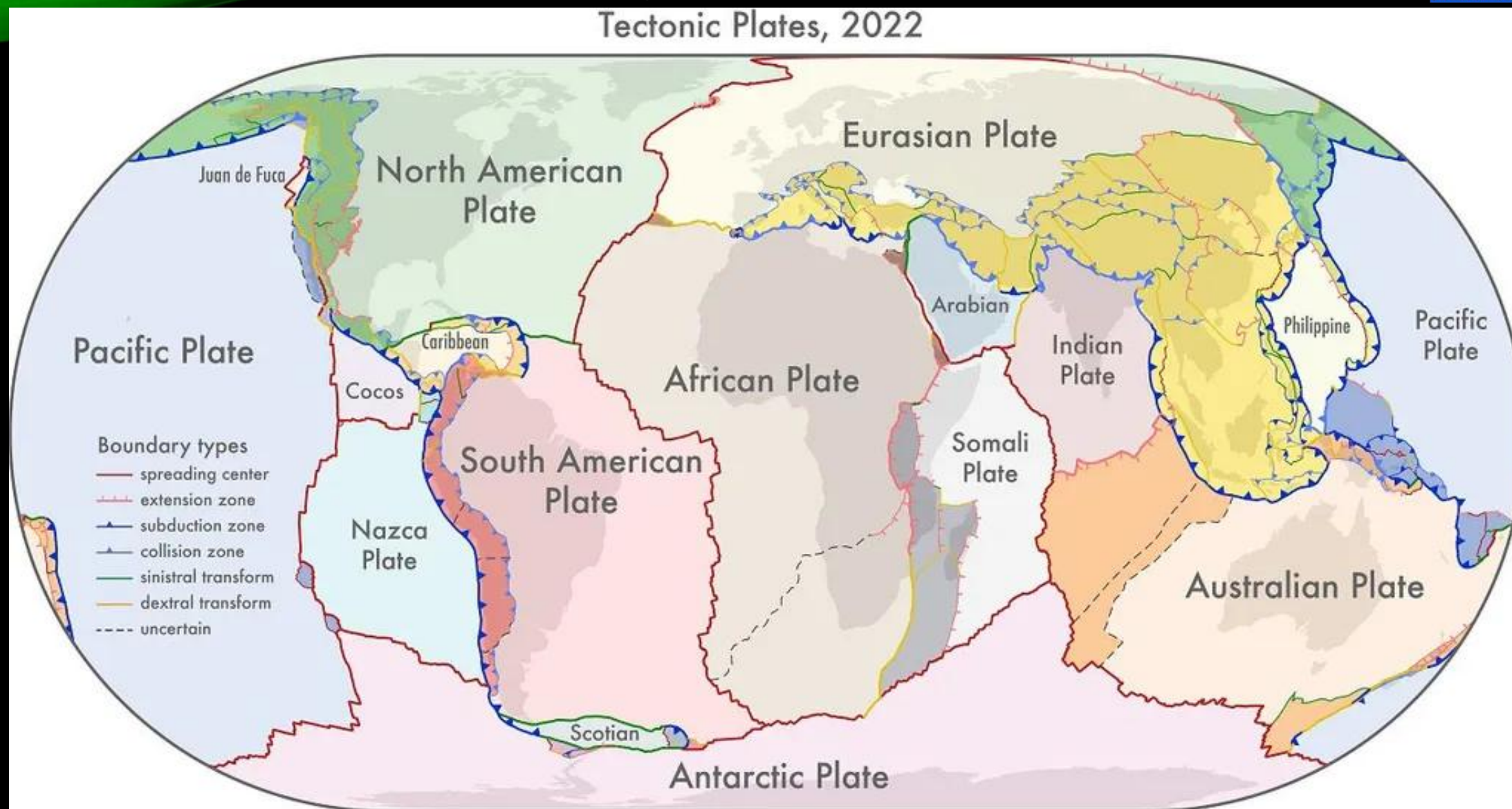
Sedimentary basins are **depressions** in the Earth's crust that are filled with sediments and vary in size from small local features to massive regional systems. They are formed by a variety of geological processes and are important for the study of Earth's history and the exploration of natural resources.



ROLE OF TECTONICS IN BASIN FORMING, EVOLUTION AND FILLING

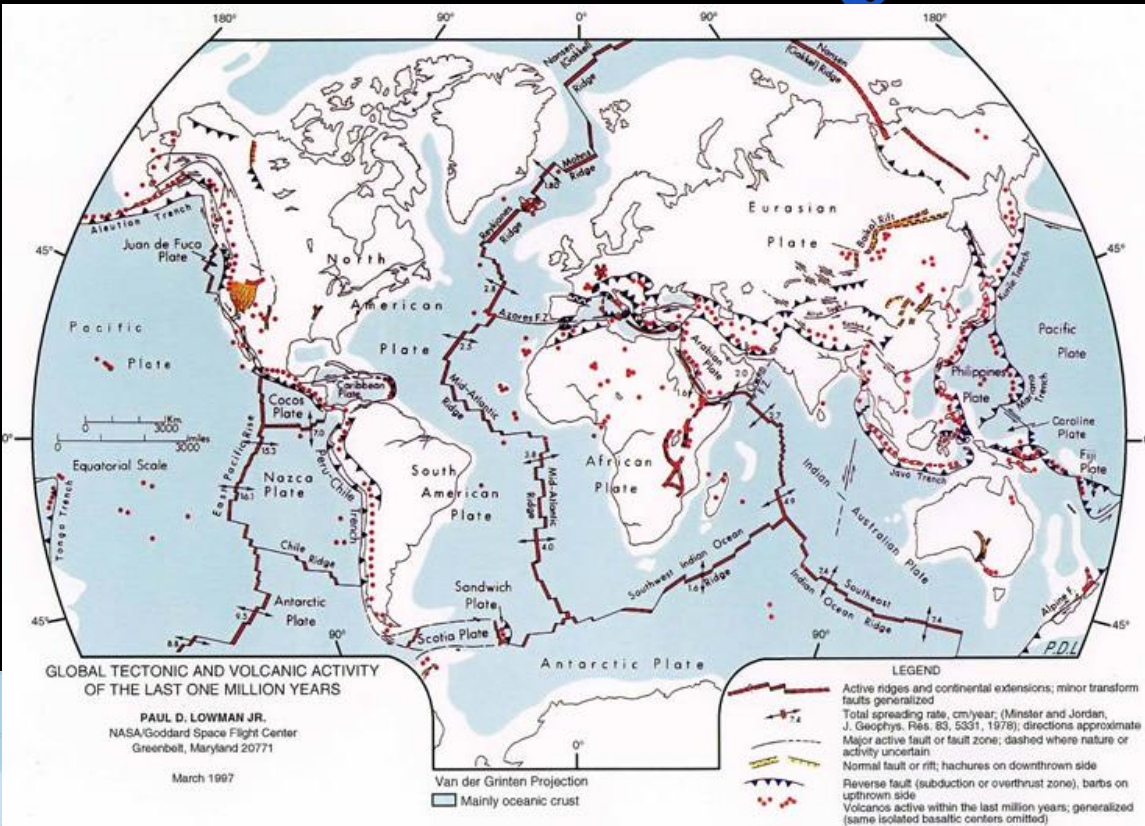
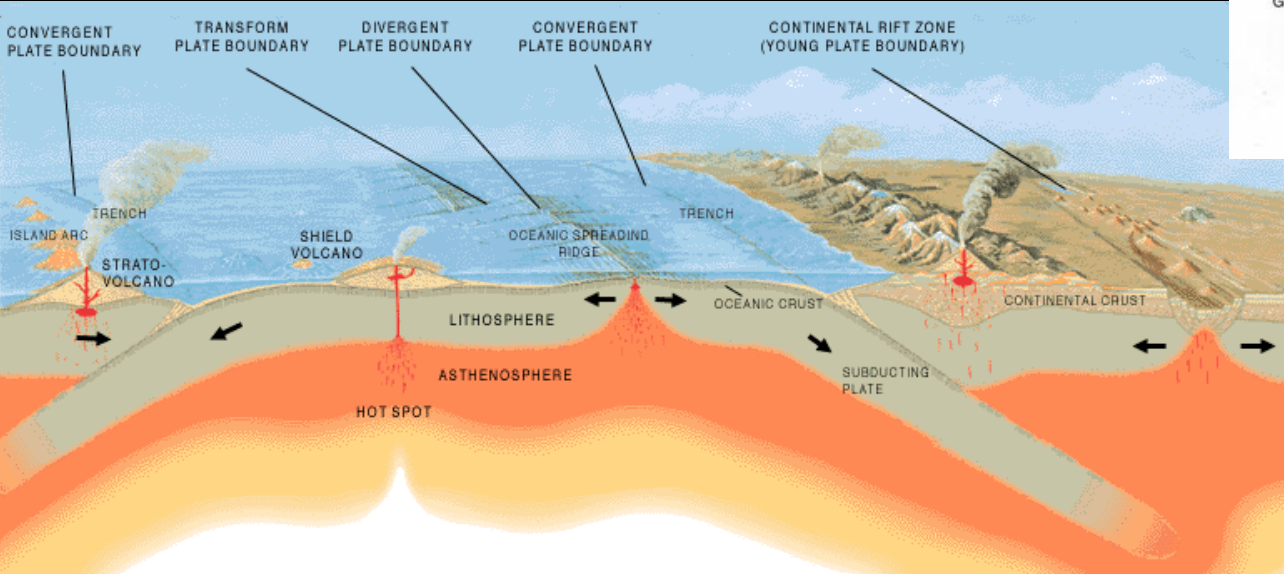
- Tectonics plays a fundamental role in the formation, evolution, and filling of sedimentary basins.
- It controls the creation of accommodation space, sediment supply, and depositional environments, which collectively determine the stratigraphic architecture of basins.

Earth's Major Tectonic Plates



Earth's Major Tectonic Plates

The outermost part of Earth's structure is known as the lithosphere. The lithosphere consists of the crust and small portion of the upper mantle. The lithosphere is divided into a number of tectonic plates. These plates move and interact with one another, driven by convectional forces within the Earth.

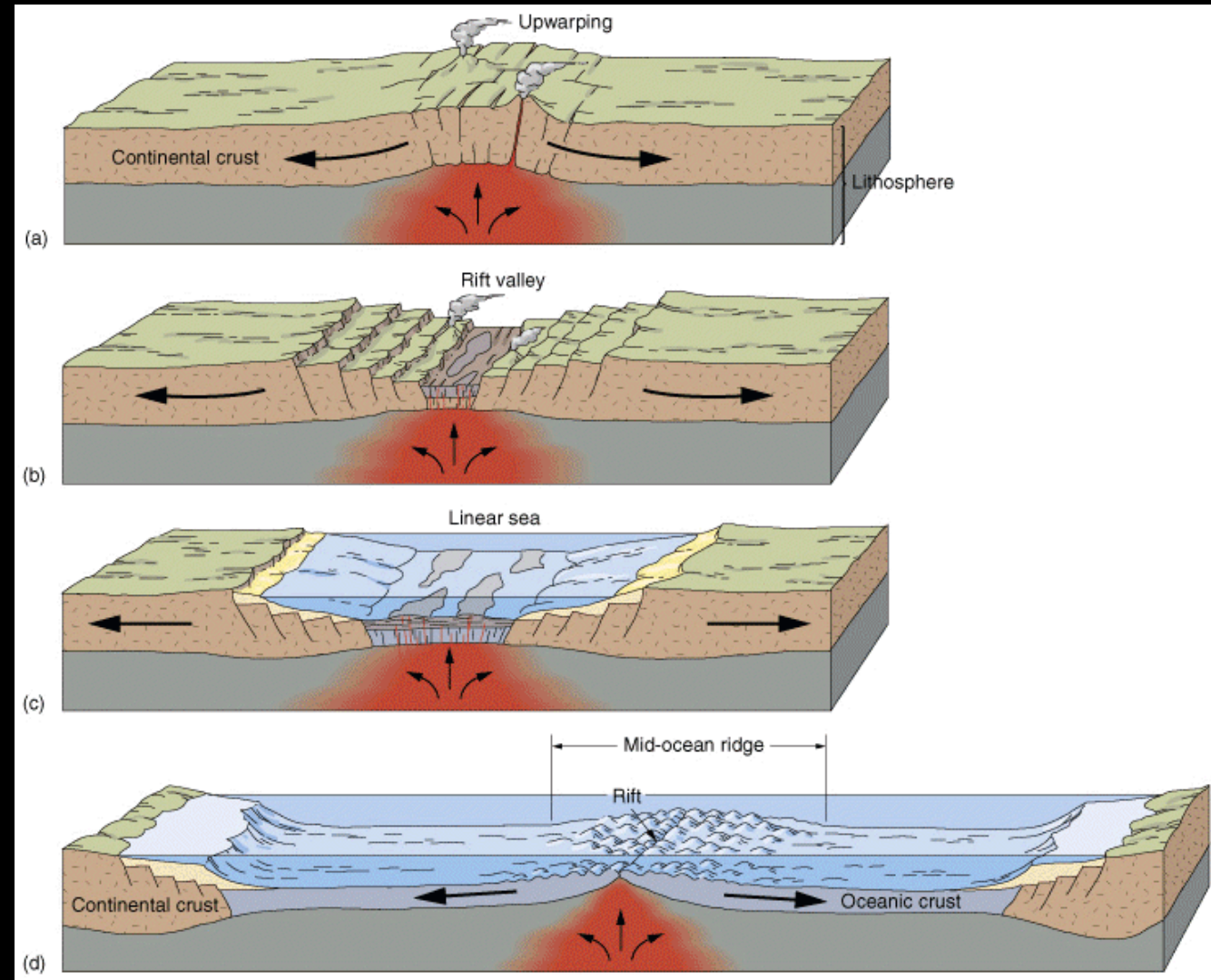


<https://geology.com/plate-tectonics.shtml>

TECTONIC SETTING AND BASIN TYPES

Types of Basins: Tectonics determines the type of basin formed, which influences the nature of basin filling. Examples include:

- **Rift Basins:** Formed by extensional tectonics (e.g., East African Rift).
- The East African Rift Basin is actively filling with sediments from rift-flank uplift and volcanic activity.





TECTONICS AND ACCOMMODATION SPACE

Accommodation Space: The space available for sediment accumulation, controlled by:

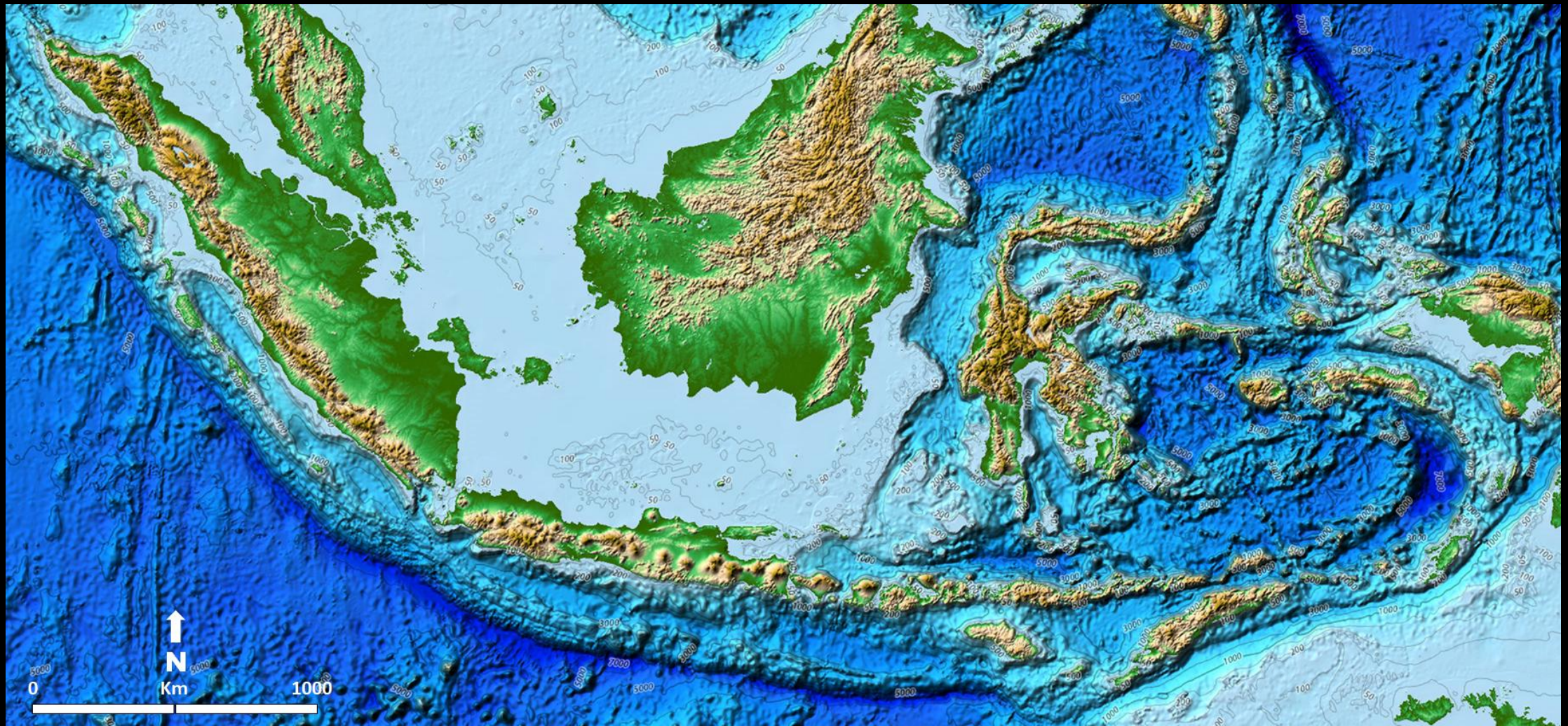
- **Subsidence:** Tectonic processes (e.g., crustal thinning, loading) create subsidence, allowing sediments to accumulate.
- **Uplift:** Tectonic uplift in source areas increases sediment supply to the basin.



TECTONICS AND SEDIMENT SUPPLY

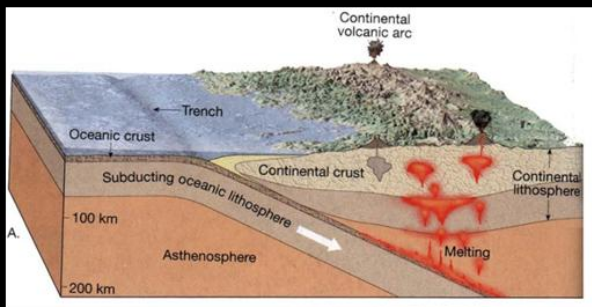
- Source Area Uplift: Tectonic uplift of mountains or highlands provides sediment to the basin.
 - Example: The Himalayas supply sediments to the Indus and Ganges foreland basins.
- Climate and Erosion: Tectonics influences climate patterns, which in turn affect erosion rates and sediment supply.
- Provenance: Tectonics determines the lithology and composition of sediments by exposing specific rock types in source areas.

Bathymetry and Shuttle Radar Topography Mission (SRTM) Map

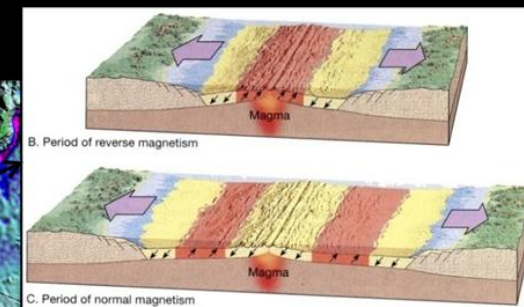


TECTONICS OF INDONESIAN REGION

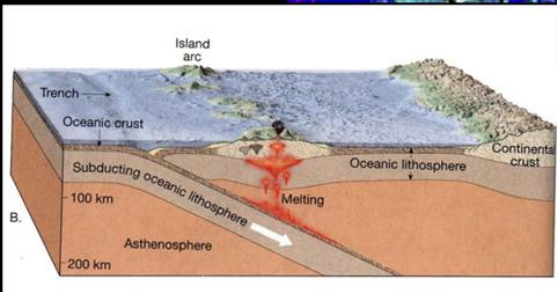
Forearc Basin



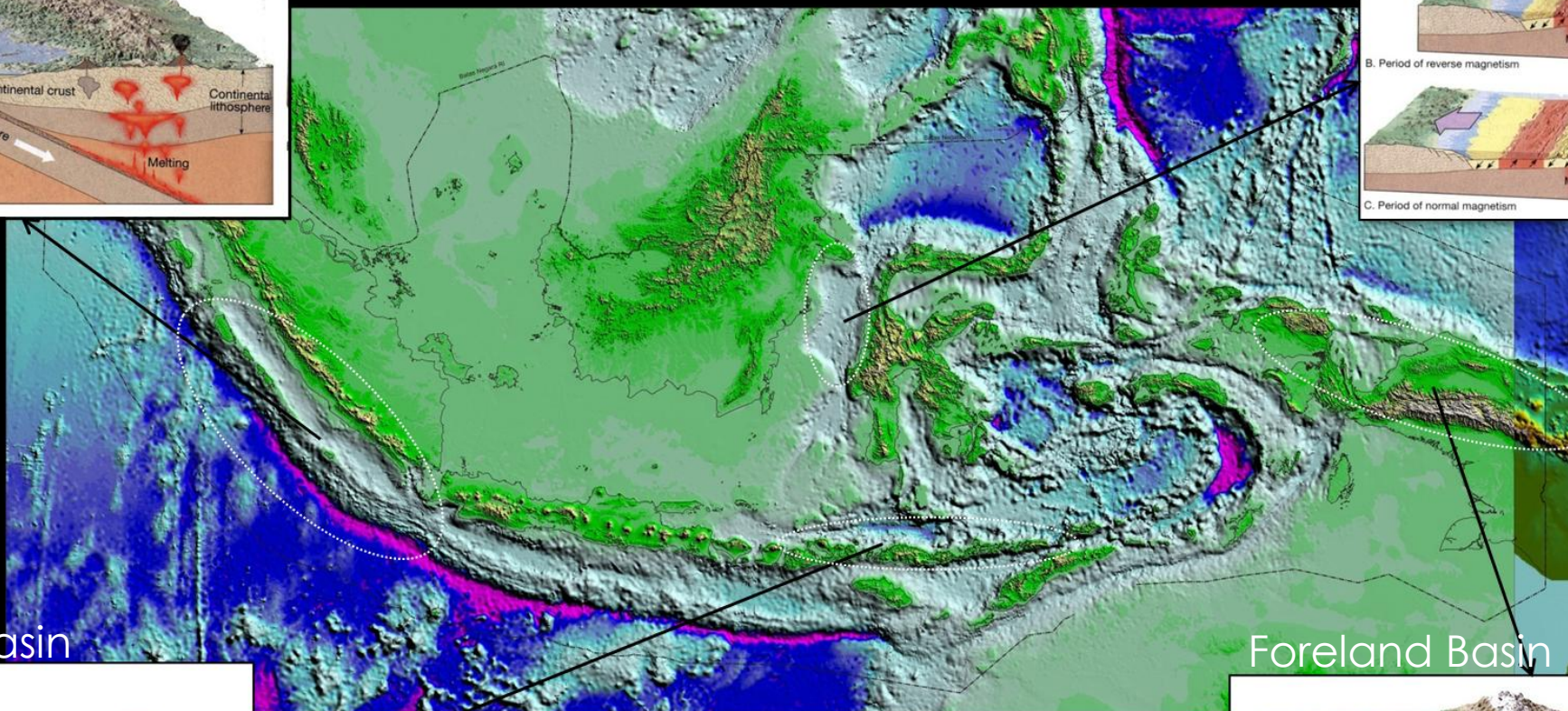
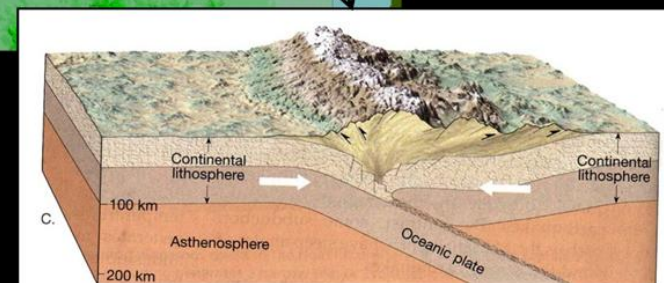
Rift Basin



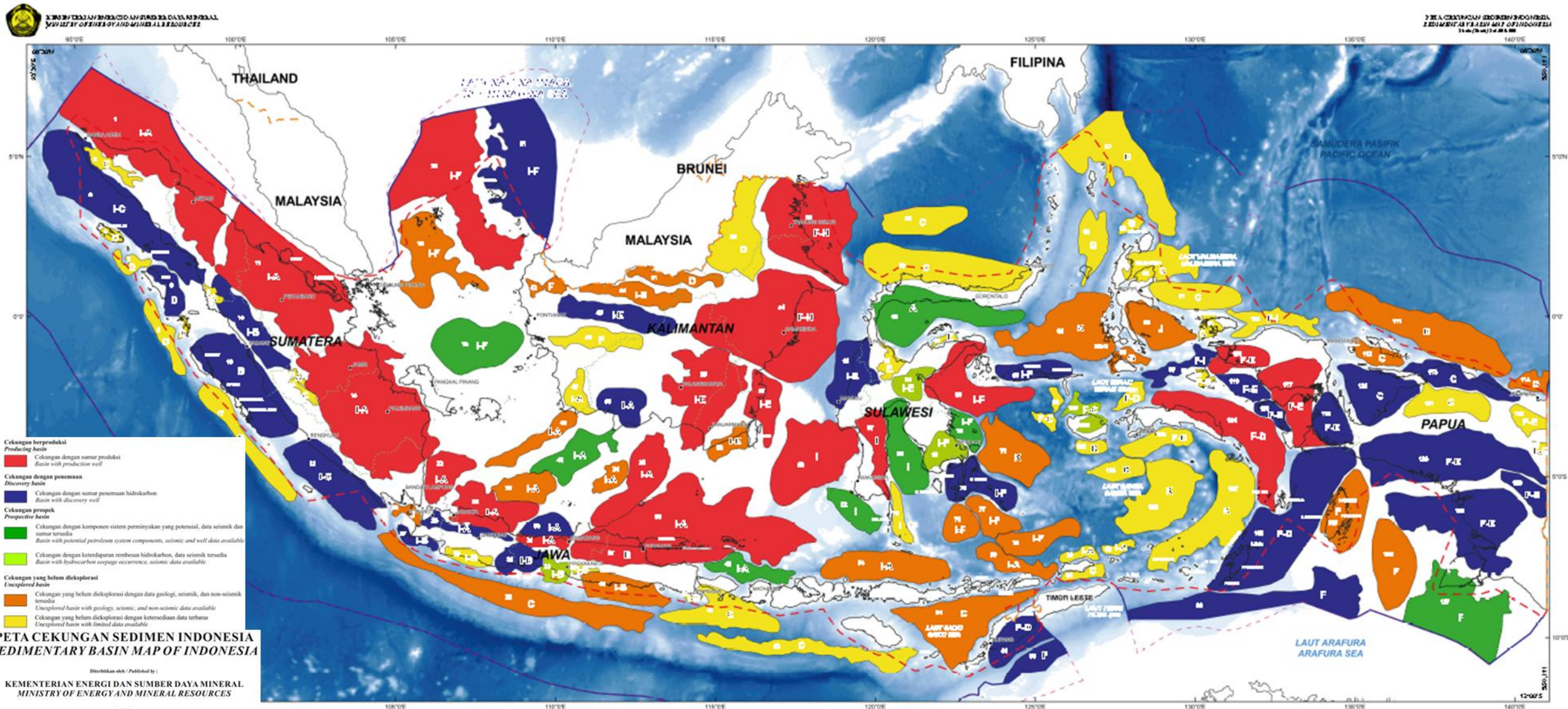
Backarc Basin

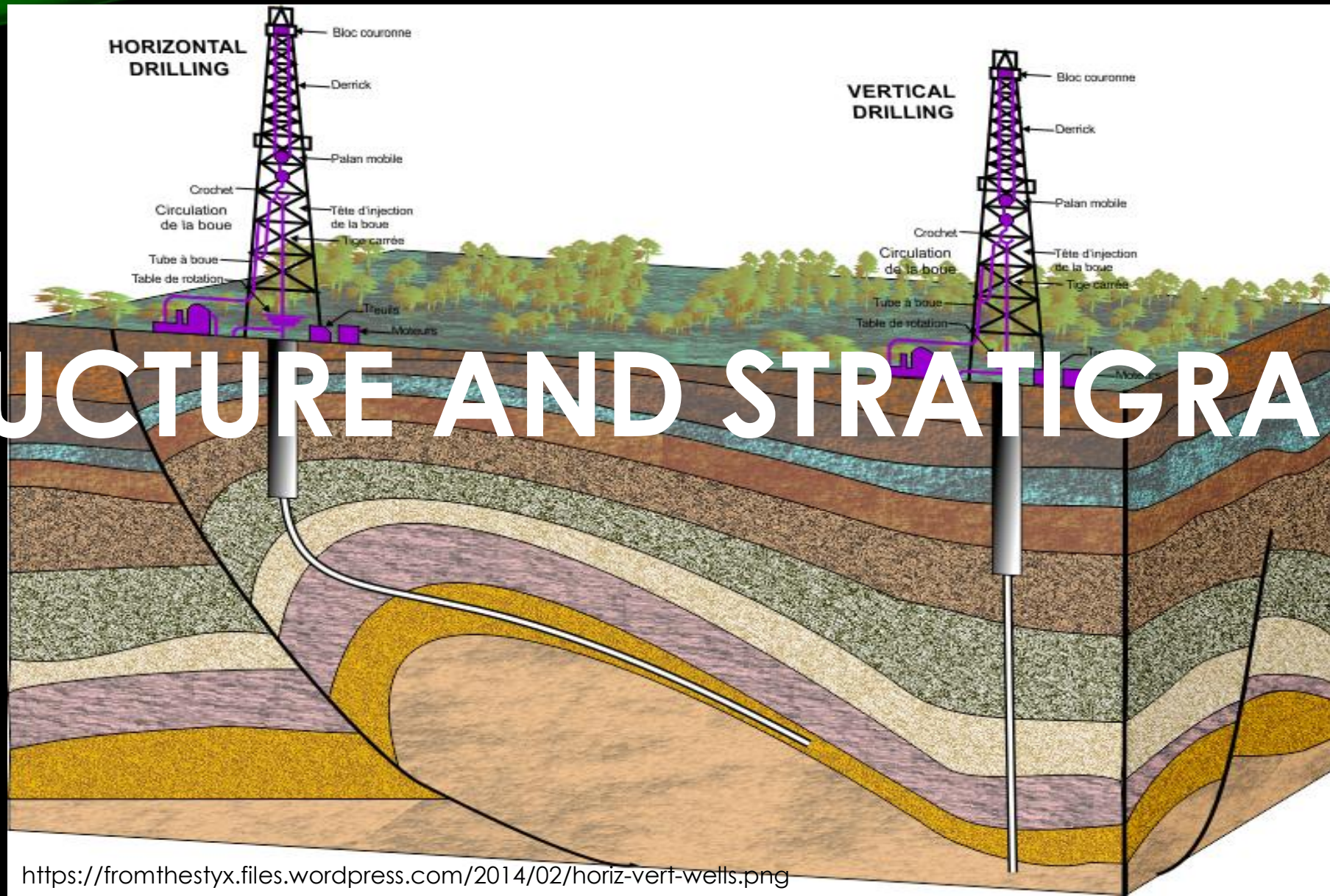


Foreland Basin



INDONESIA TERTIARY BASIN (ESDM, 2022)

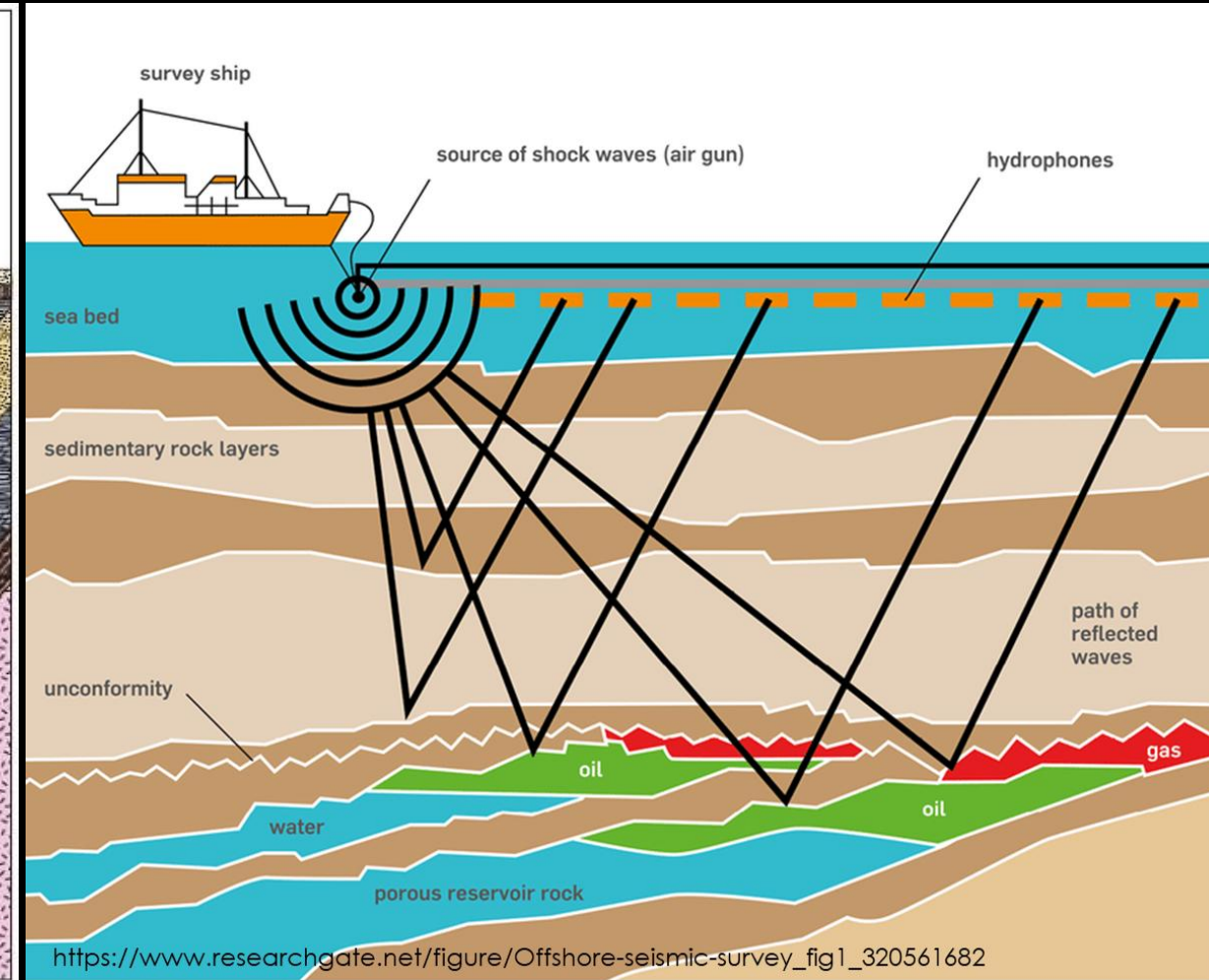
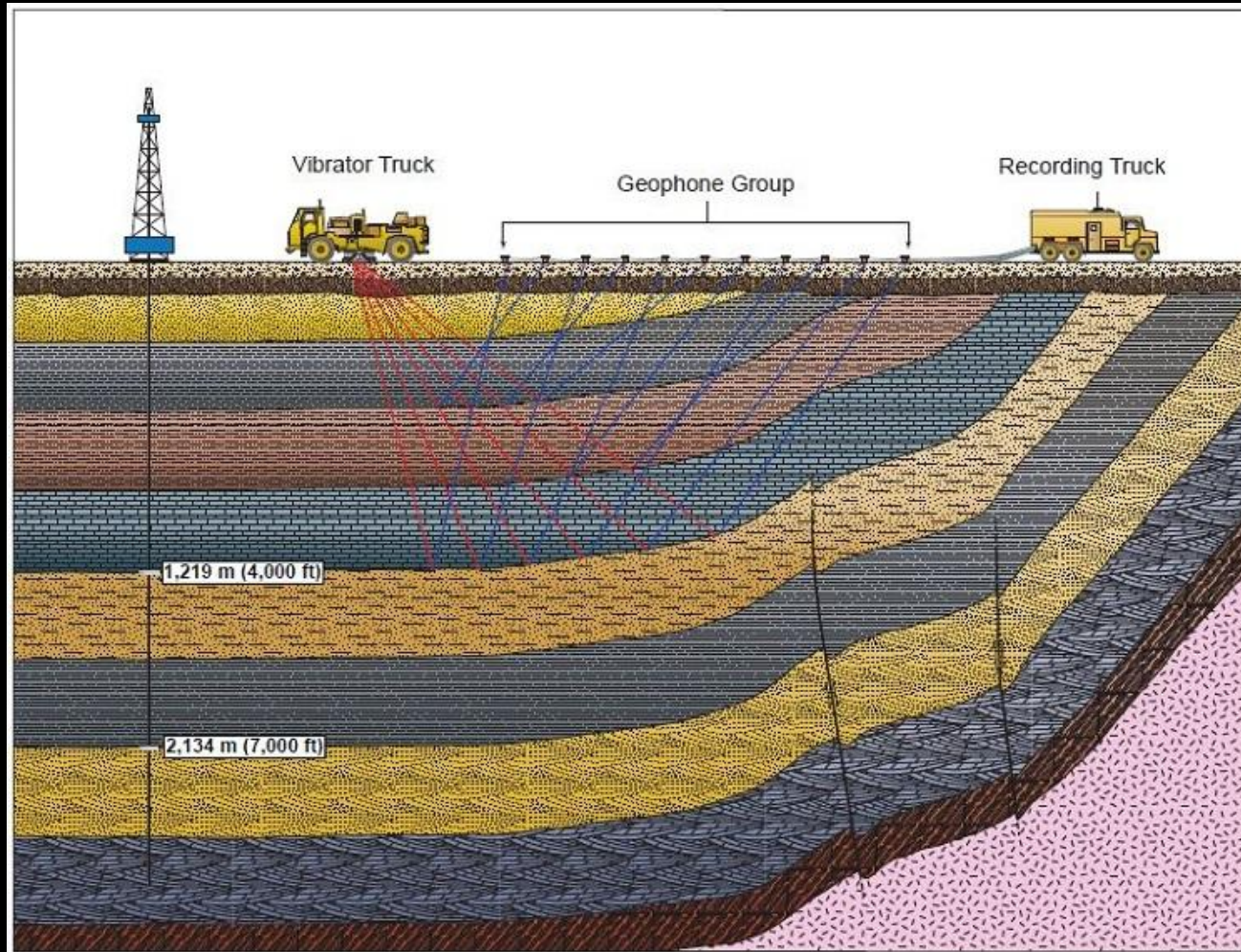




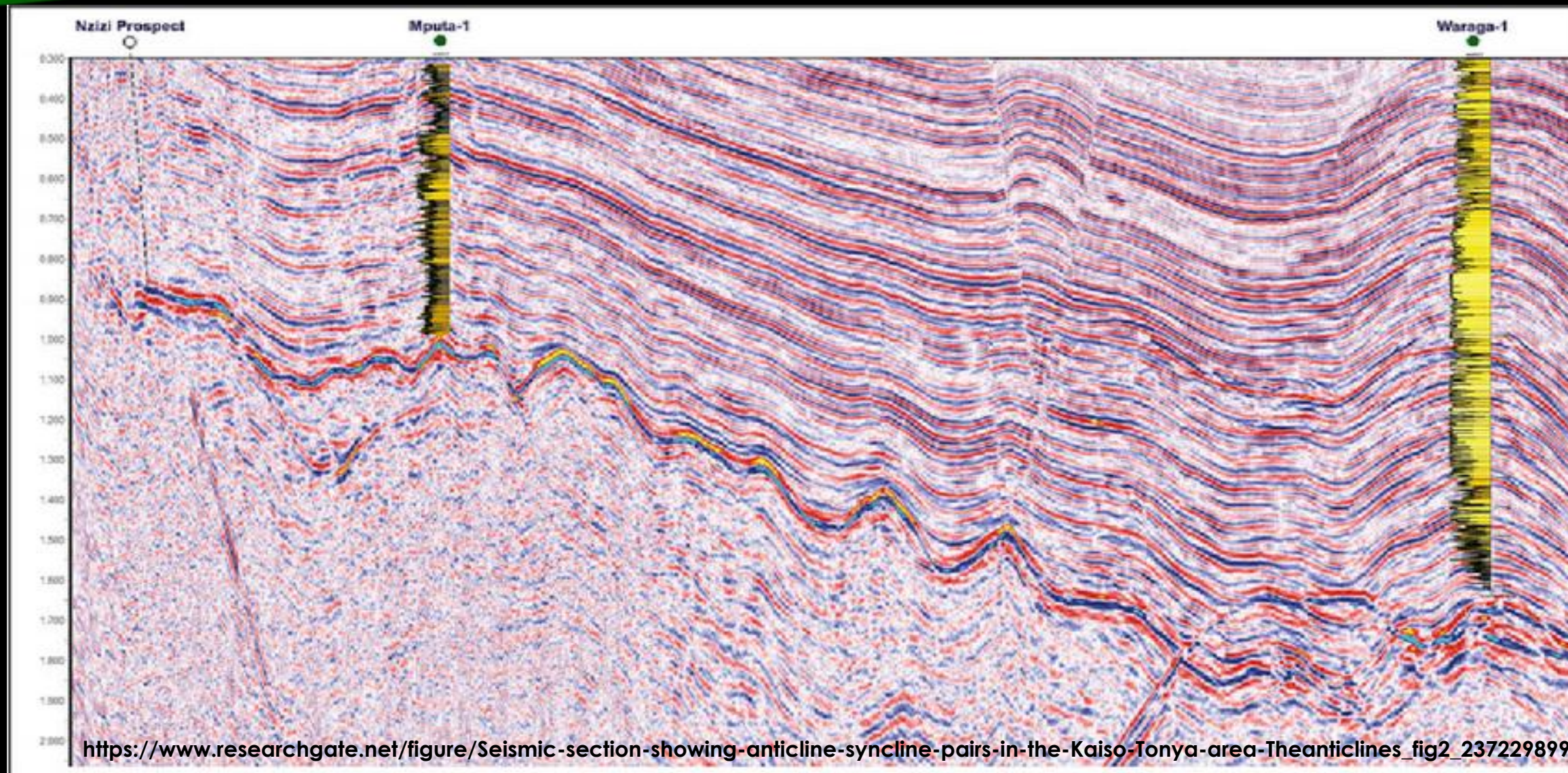
STRUCTURE AND STRATIGRAPHY

SEISMIC ACQUISITION

Onshore Vs Offshore



SUBSURFACE STRUCTURAL CONFIGURATION



SURFACE GEOLOGY



Oil Seep





PETROLEUM SYSTEM

- **Generative sub-system**

Element : source rocks
Processes : Thermal maturation

- **Migration and Entrapment Sub-system**

Migration

Element : Generated hydrocarbon
Processes : Expulsion (Primary migration): Source rock to Carrier bed
Secondary migration: Within carrier bed to traps

Entrapment Sub-system

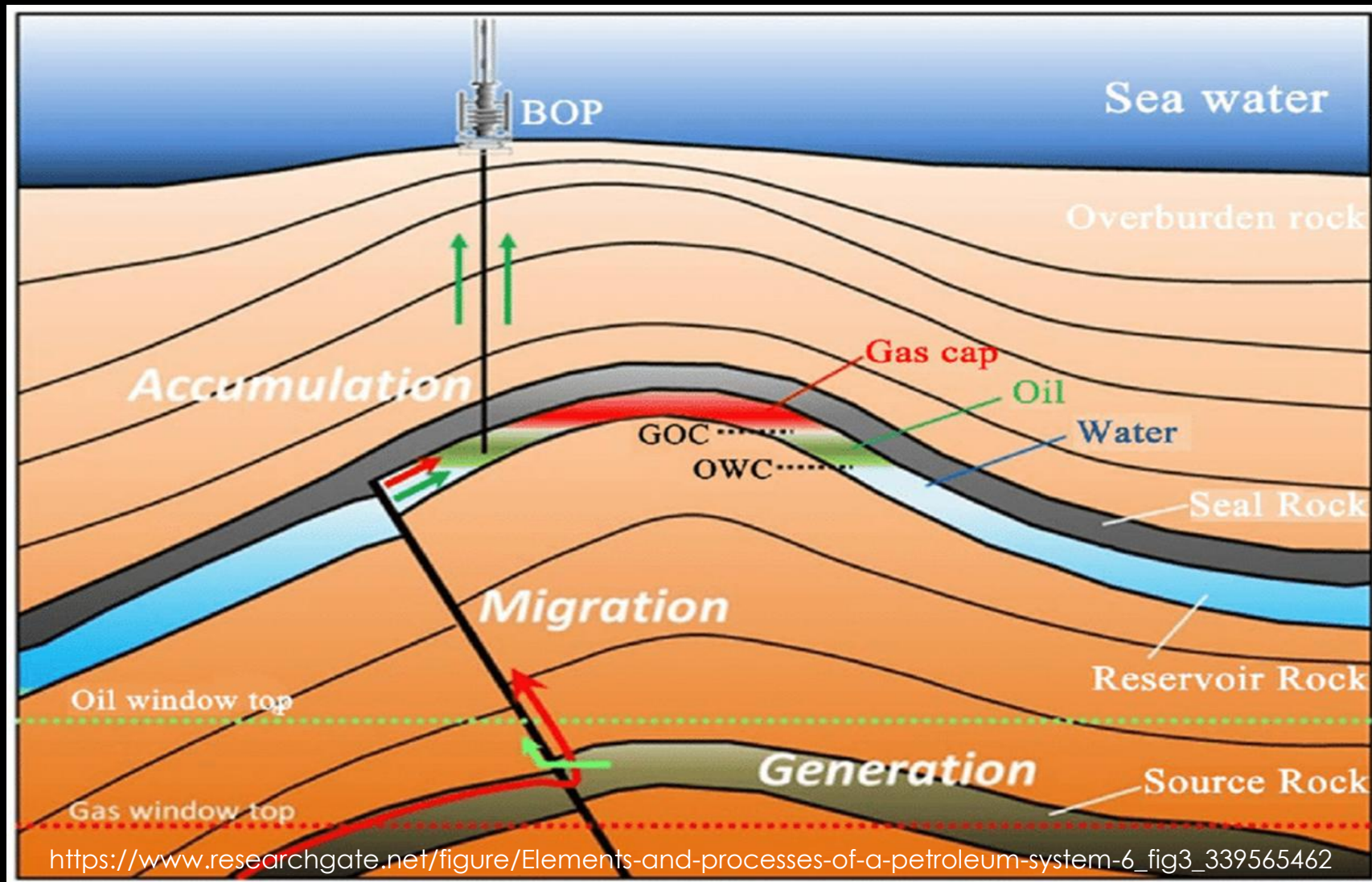
Element :

- Migrated hydrocarbon
- Trap geometry
- Reservoir rocks
- Seal rock

Processes :

- Migration
- Trapping

ELEMENTS AND PROCESSES OF A PETROLEUM SYSTEM





HYDROCARBON DEFINITION

Organic chemical compounds of hydrogen and carbon atoms that form the basis of all petroleum products. They may exist as solids, liquids or gases.

([www. Conocophillips.com](http://www.Conocophillips.com))



SOURCE ROCK

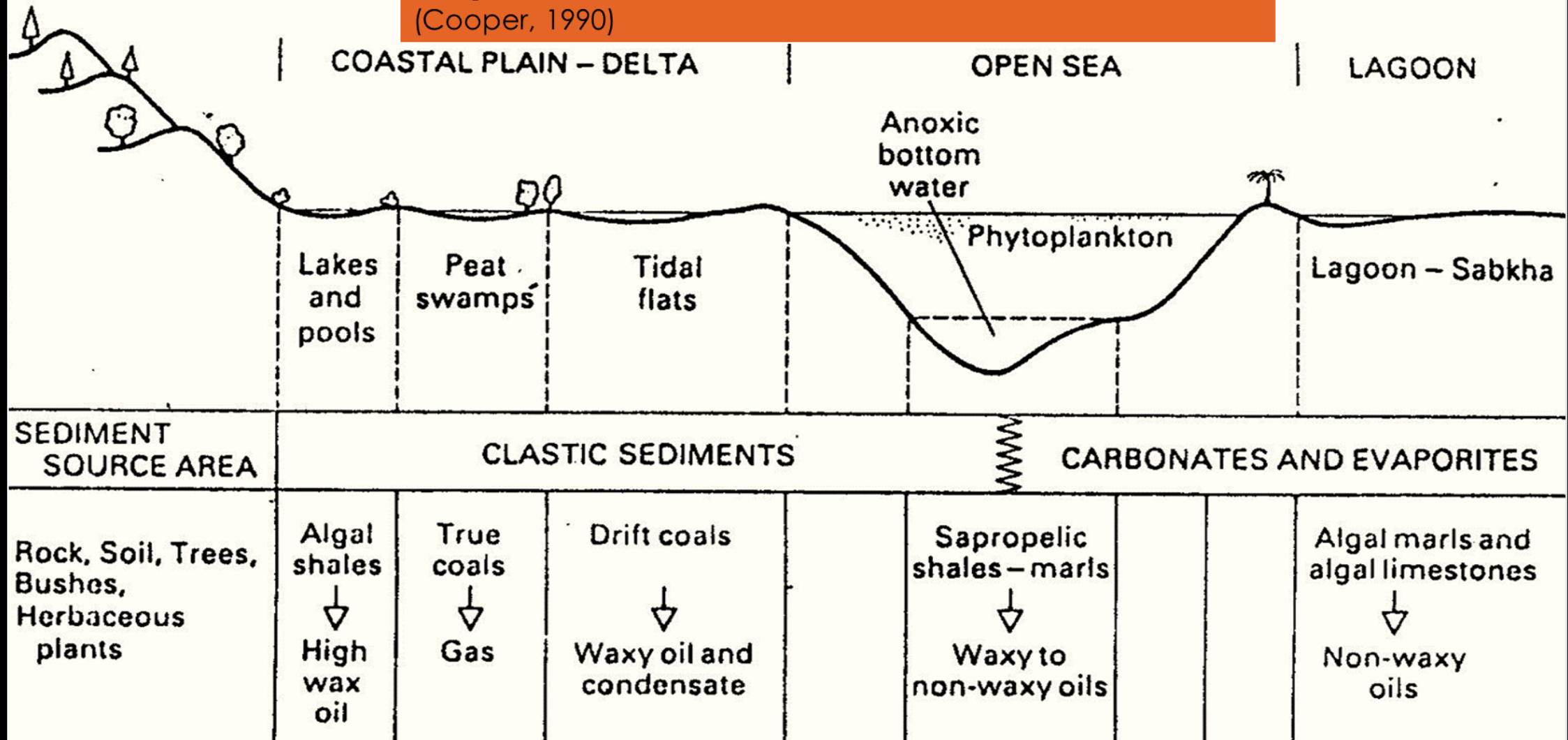
Organic Theory

- Worldwide observations by almost all petroleum geologists confirm a genetic correlation between oil and gas and organic-rich sediments known as source rocks.
- In addition, a large number of complex molecular structures are present in organic matter that belongs to the originating sediments without major changes of known biological molecules.

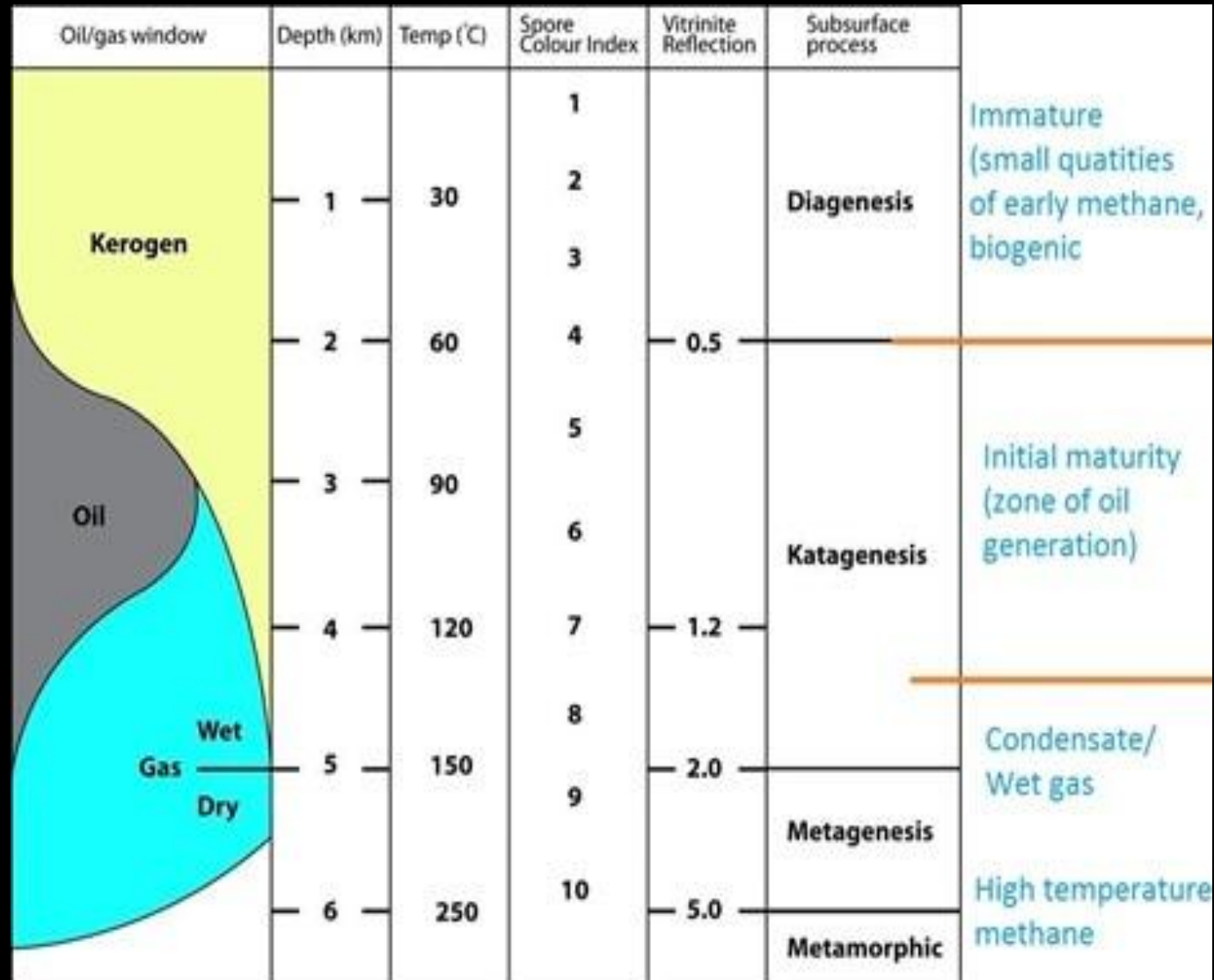
SOURCE ROCK

Depositional environment of source rocks

(Cooper, 1990)

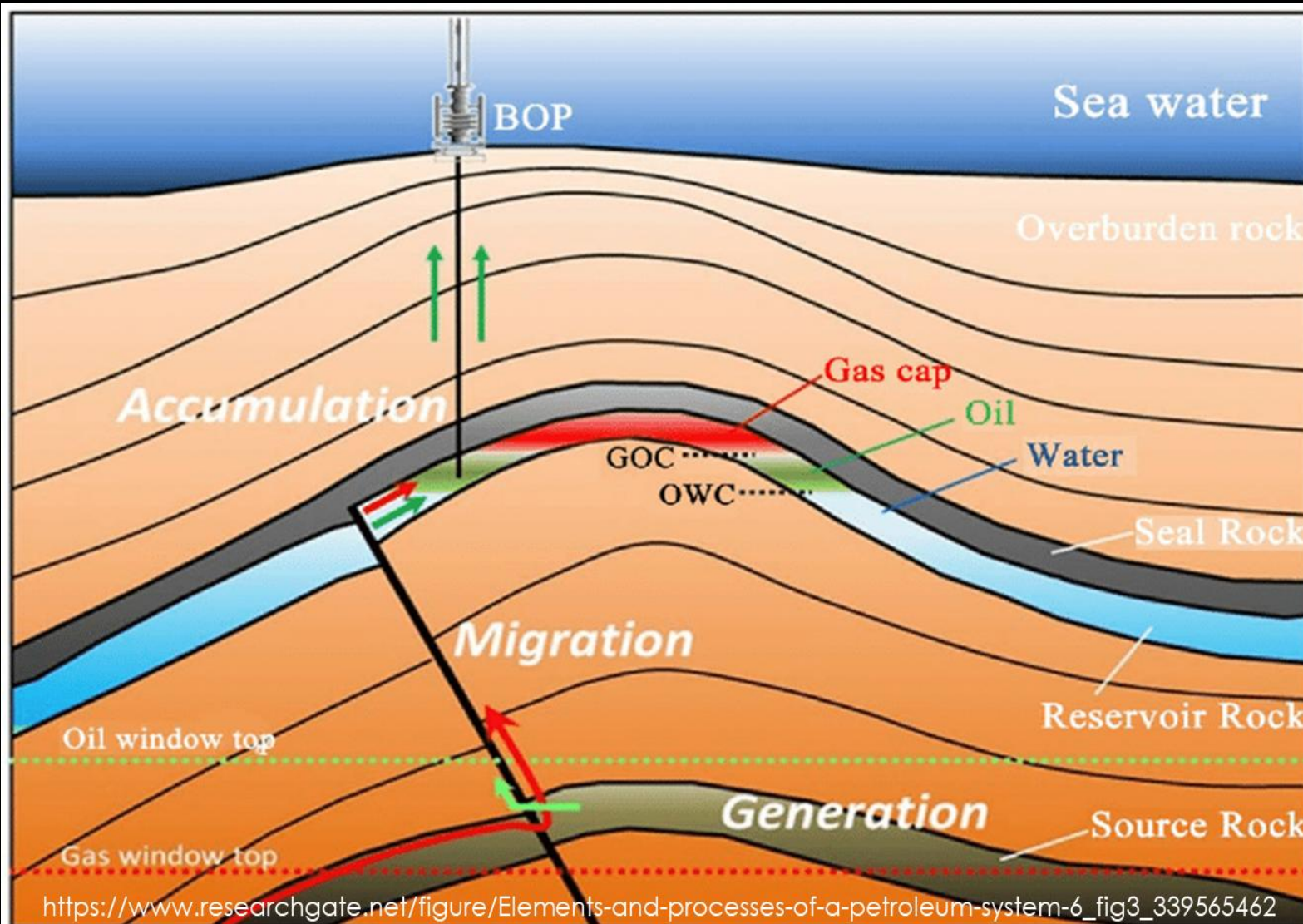


SOURCE ROCK MATURATION



This figure shows each type of maturation processes along depth (km).

HYDROCARBON MIGRATION



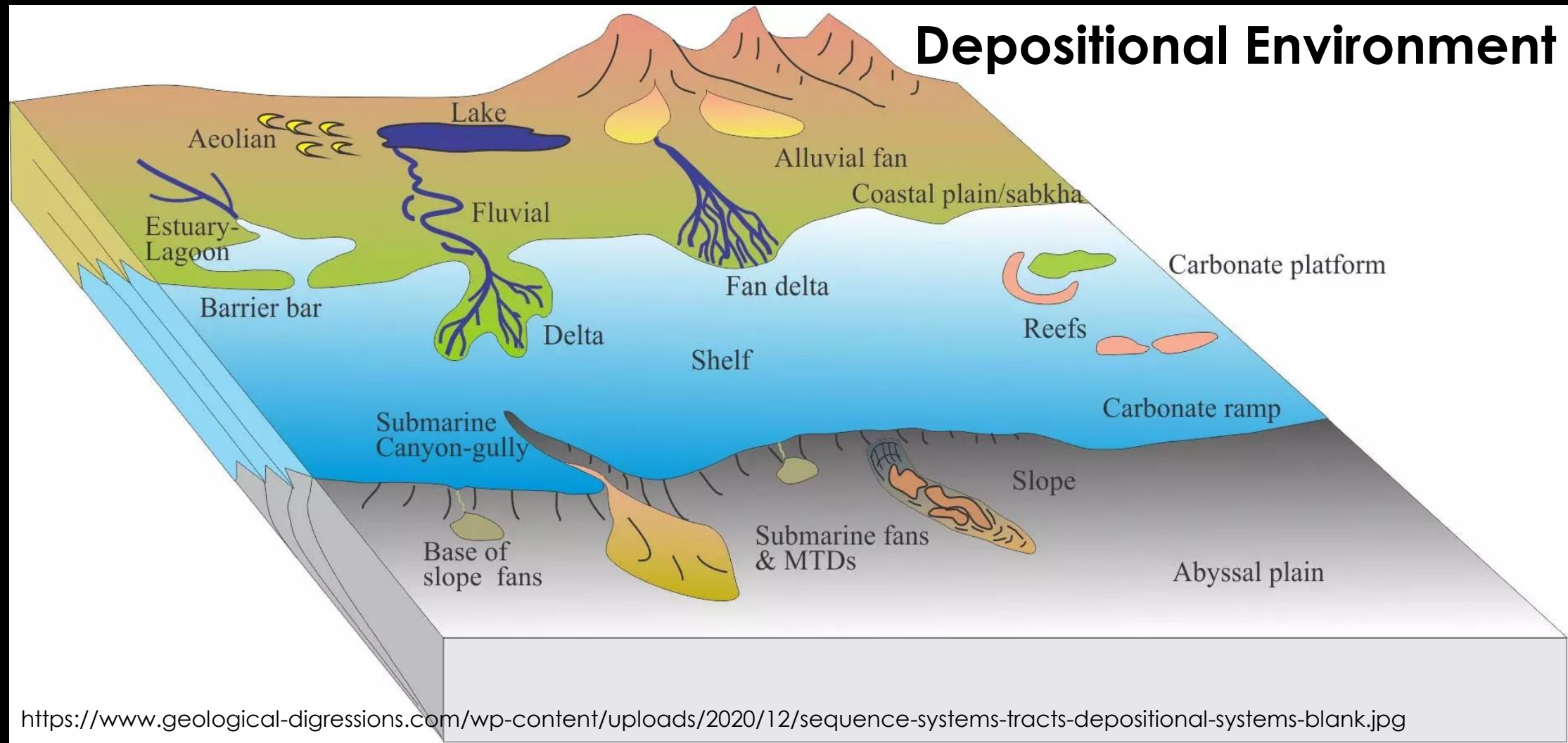
Hydrocarbon Migration

- Expulsion (Primary migration): Source rock to Carrier bed
- Secondary migration: Within carrier bed to traps



Reservoirs

PETROLEUM SYSTEM – RESERVOIRS



PETROLEUM SYSTEM – RESERVOIRS

Siliceous;
 Silicon-
 bearing Clasts;
 broken-up pieces
Siliciclastics generated from pre-existing rocks

Sandstone
 Conglomerate
 Shale

Biological generated by organisms

Limestone
 Chalk
 Chert
 Coal

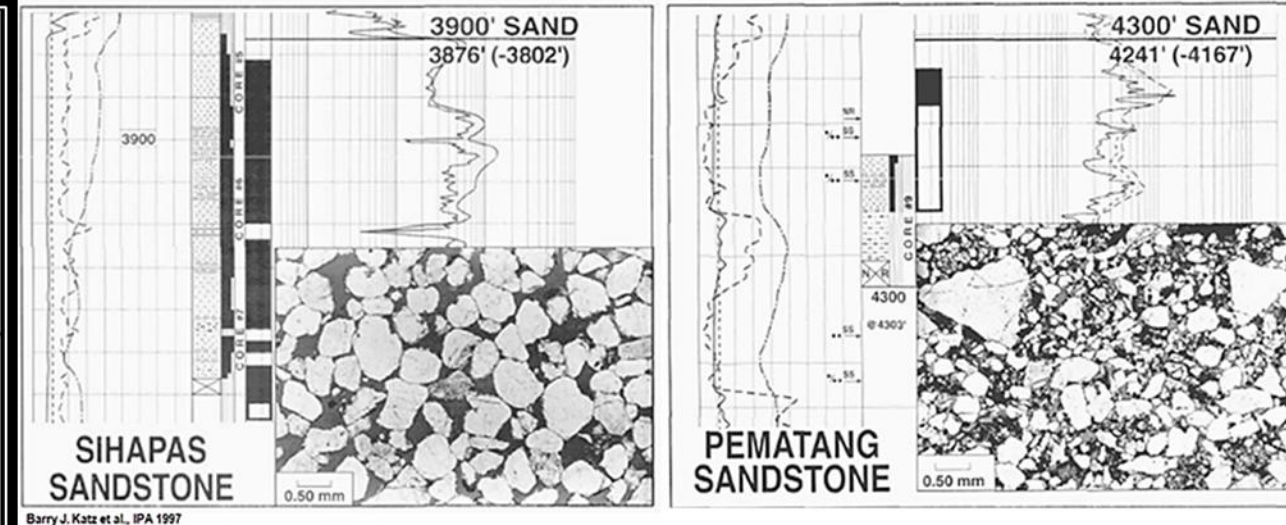
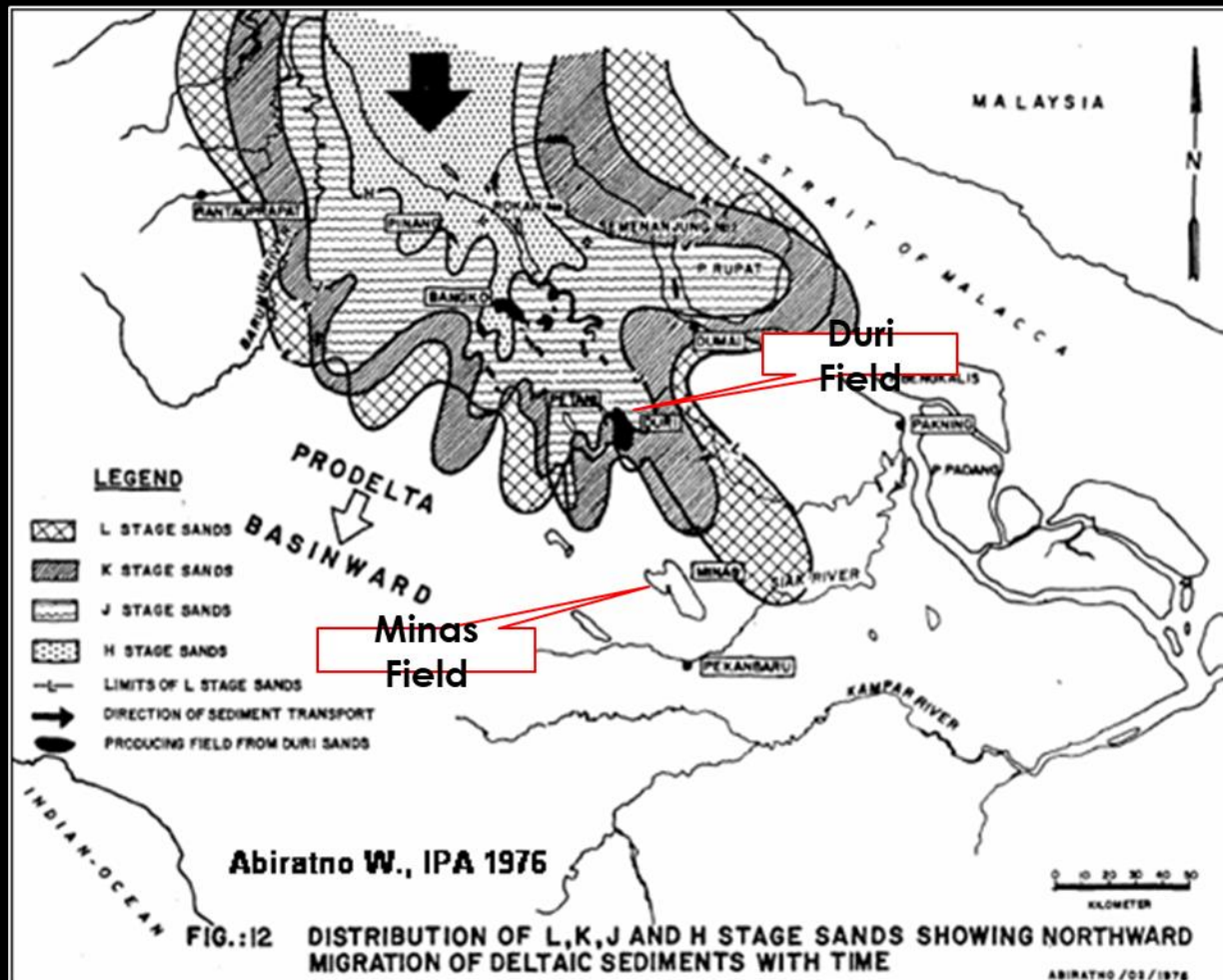
Chemical generated from dissolved solids

Rock salt (evaporites)

Types of Sediments and Sedimentary Rocks

PETROLEUM SYSTEM – RESERVOIRS

Depositional Setting of Primary Reservoir – Central Sumatra Basin



Sihapas Group :

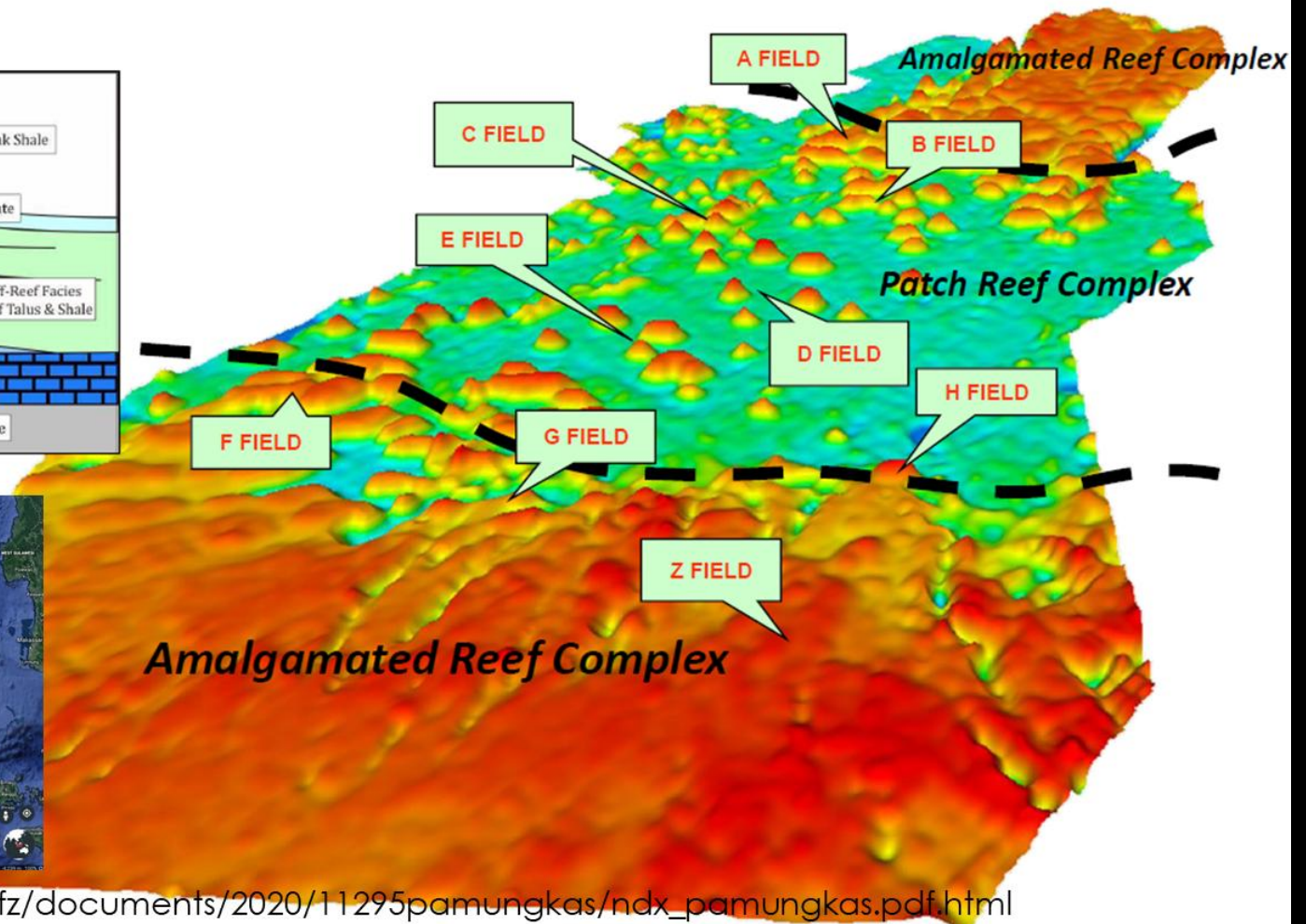
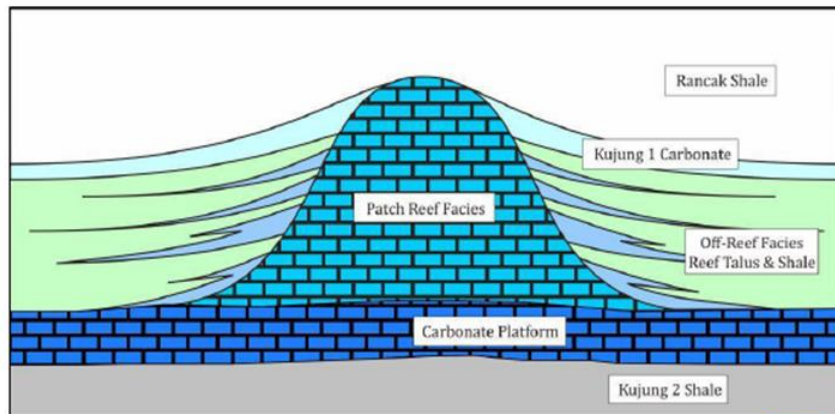
- > 90 % in-place reserves comes from Sihapas Sandstone
- Upper Sihapas sandstone is primary producer for Minas and Duri Fields
- Upper sihapas sands were mainly deposited in tidally dominated delta complex

Pematang Group :

- Pematang sands consists of alluvial-fluvial, lacustrine deltaic and fanglomerate sandstone.

PETROLEUM SYSTEM – RESERVOIRS

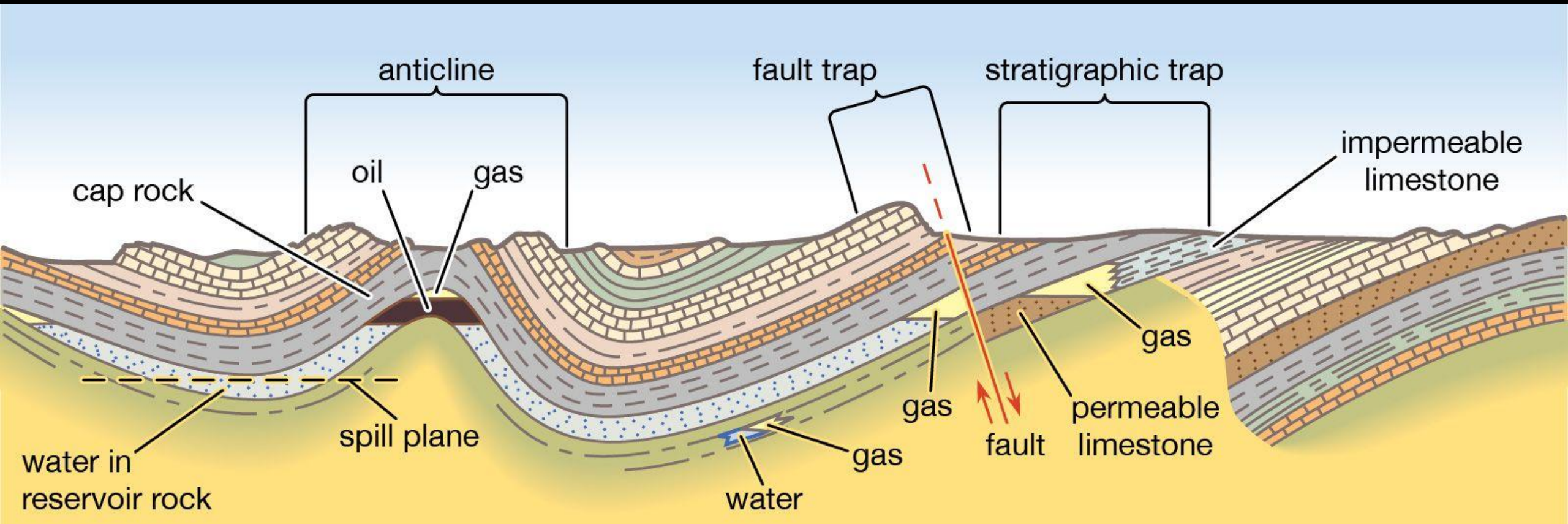
CARBONATE BUILD-UP - KUJUNG FORMATION



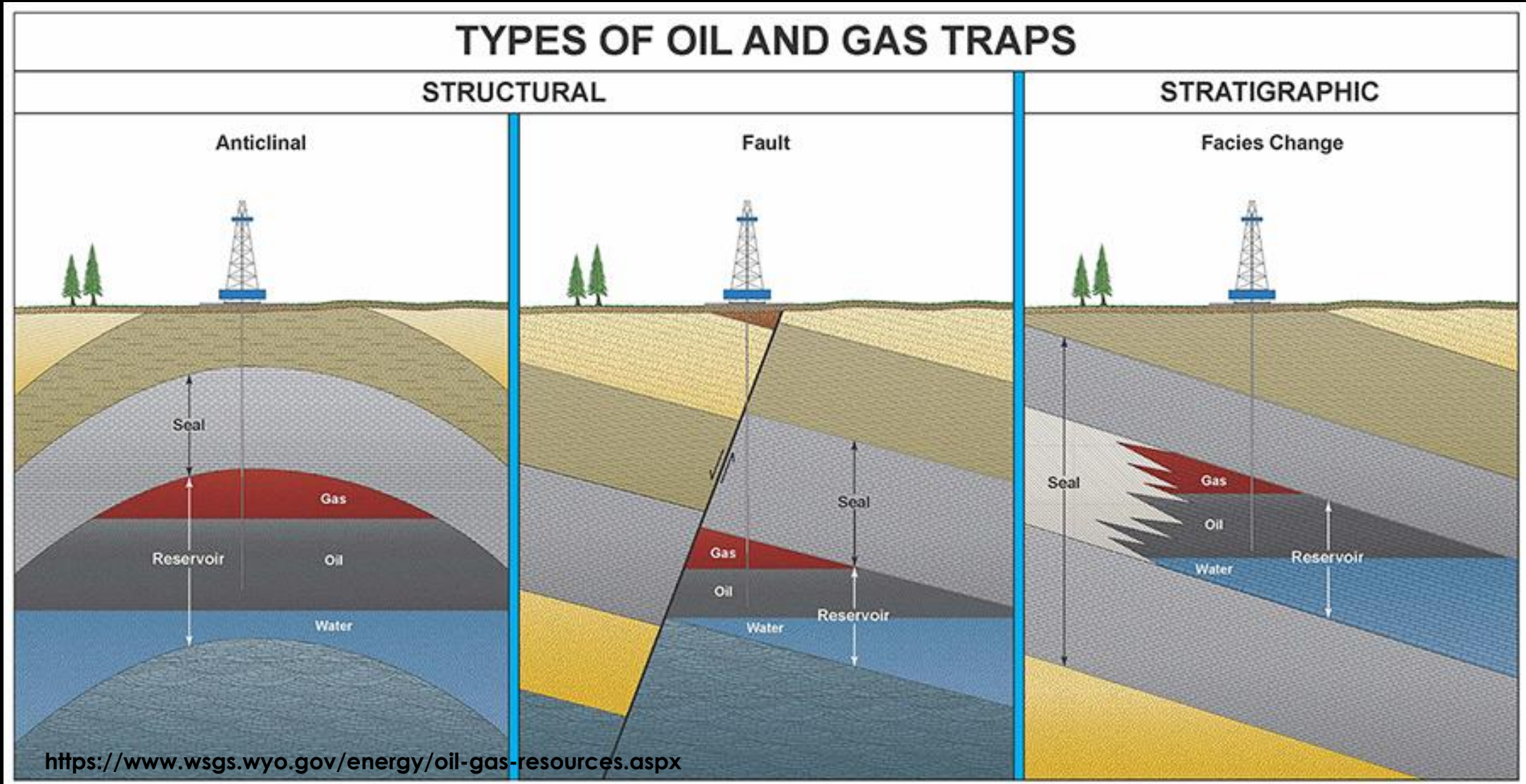


Traps and Seals

PETROLEUM SYSTEM – TRAPS AND SEALS



PETROLEUM SYSTEM – TRAPS AND SEALS



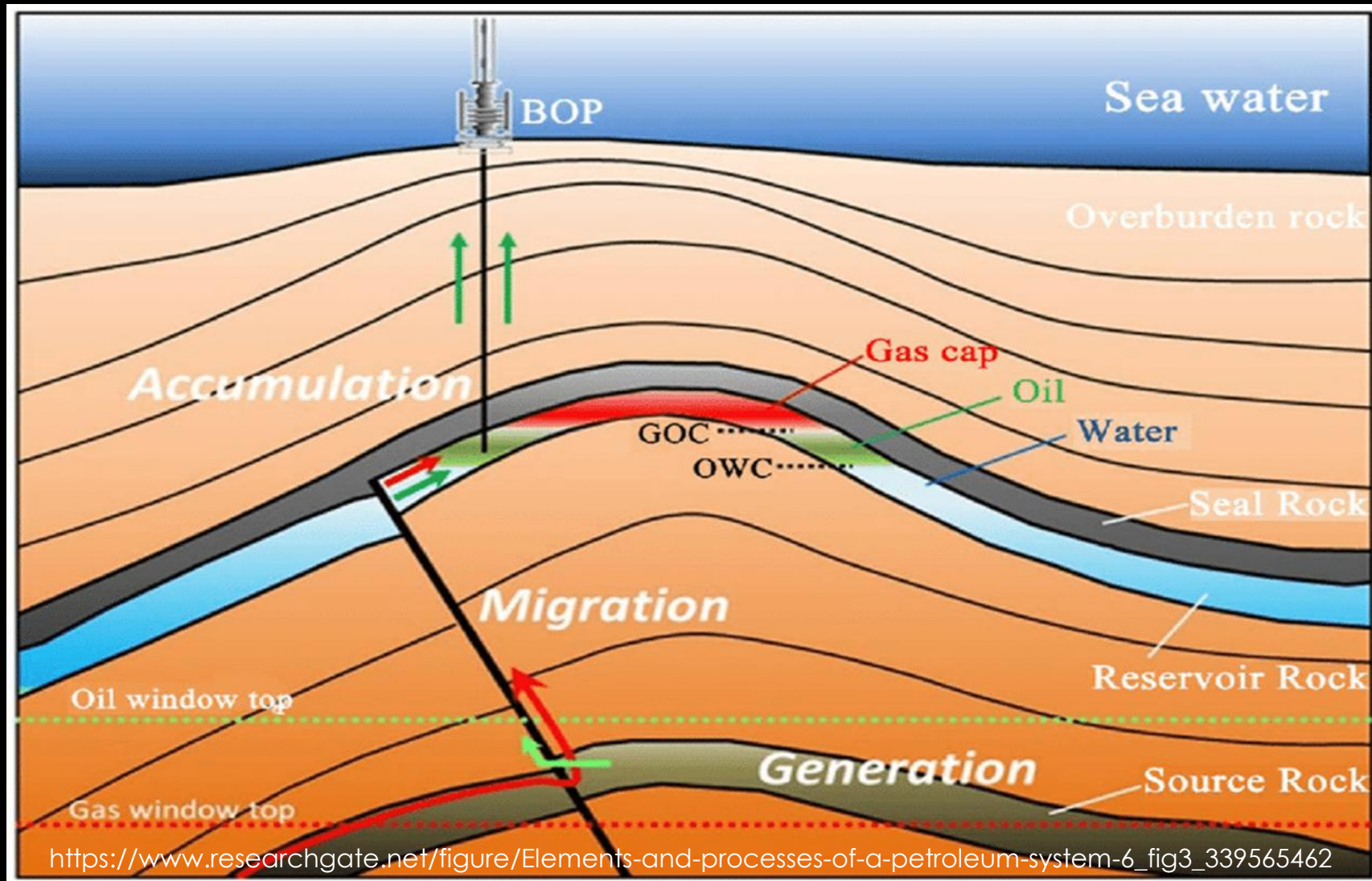


PETROLEUM SYSTEM – TRAPS AND SEALS

What is a Seal?

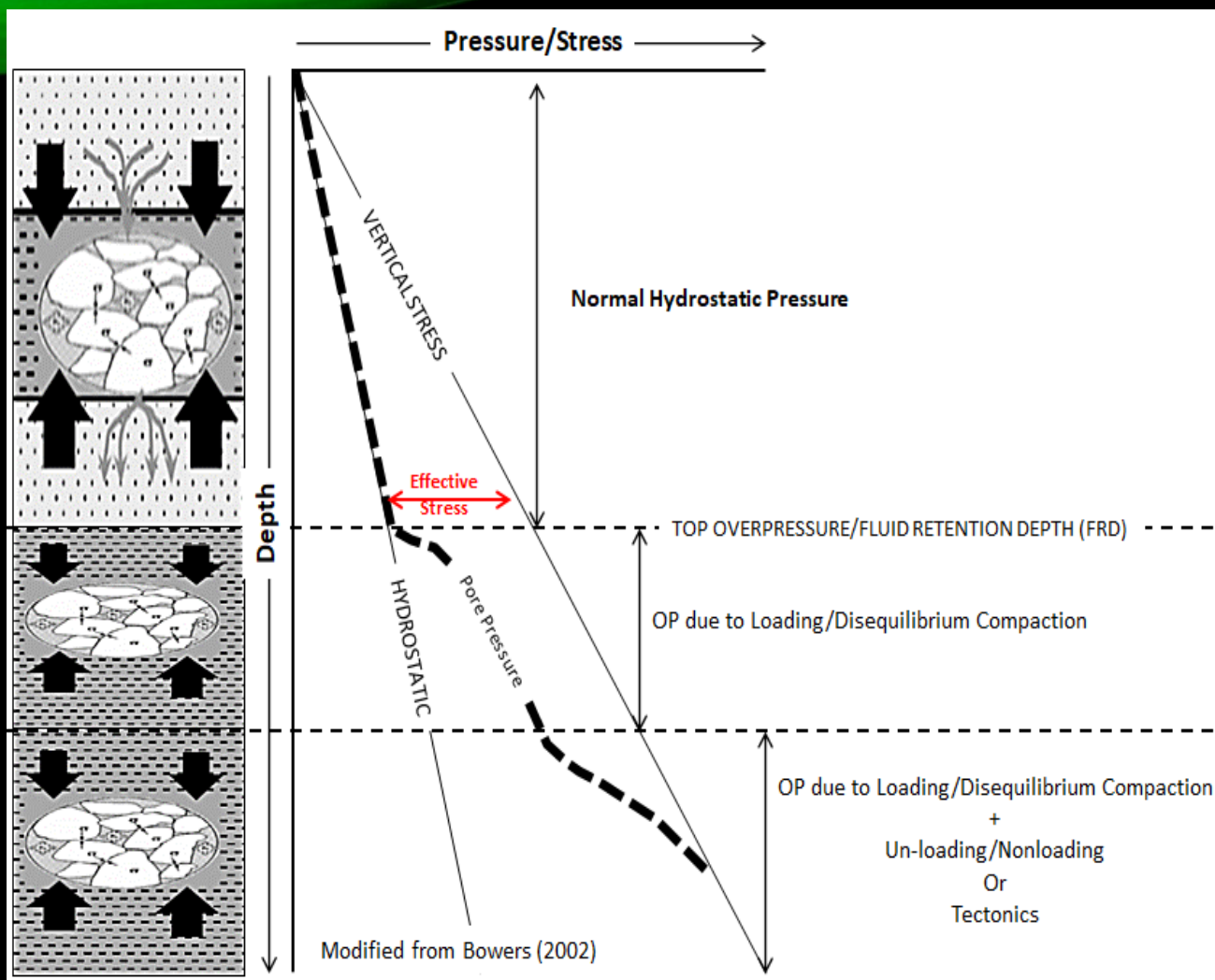
- A seal is a relatively impermeable, a seal is a layer of rock that forms a barrier or cap above and around a reservoir rock.
- Commonly composed of shale, chinks, clays, anhydrite or salt, a seal helps prevent fluids from migrating beyond the reservoir.
- It is sometimes also referred to as a cap rock.
- Rock seals can vary from layers of a few feet thick, to several hundreds of feet thick.
- Although thicker seals are usually more effective for hydrocarbon accumulation, if a seal has low permeability, is ductile, laterally continuous, and has high capillary entry pressures it will still be very effective regardless of its thickness

ELEMENTS AND PROCESSES OF A PETROLEUM SYSTEM





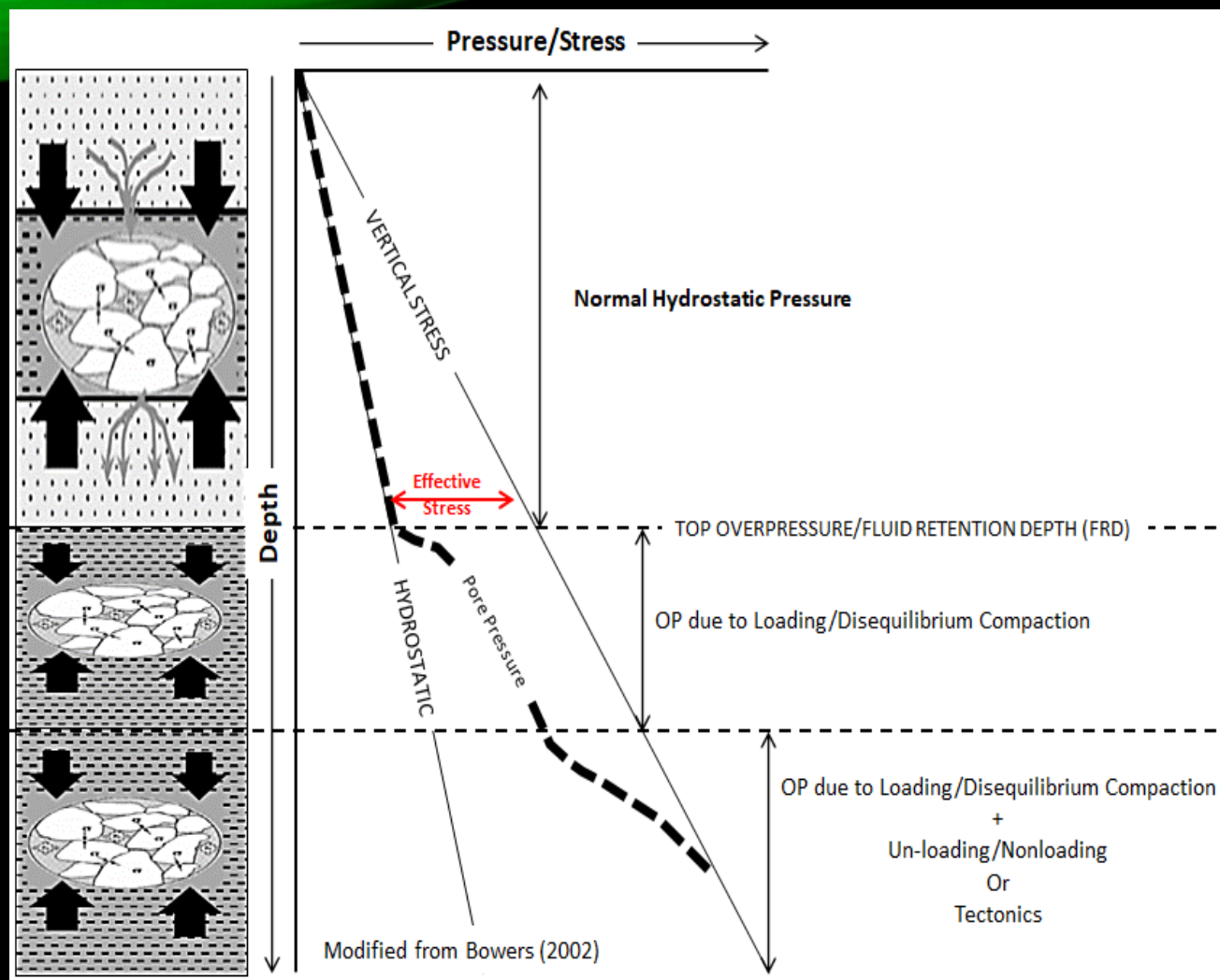
PRESSURE AND STRESS - BASIC CONCEPT



Hydrostatic pressure illustration



Pore pressure, often referred to as formation pressure, is the pressure exerted by fluids such as water, oil, or gas contained within the pore spaces of subsurface rock formations.

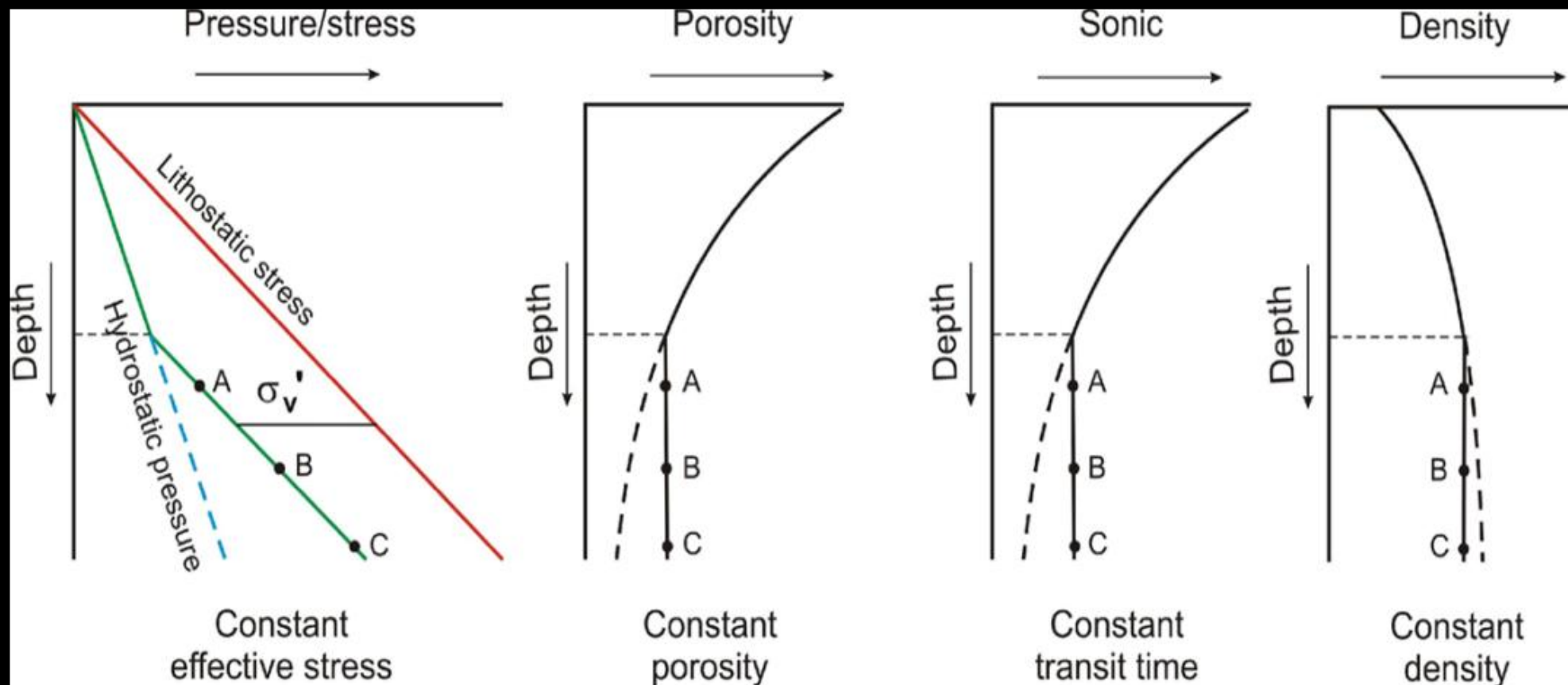


Vertical stress illustration



Vertical stress (Overburden, Lithostatic) is the stress exerted by the weight of the overlying rock column at a given depth.

OVERPRESSURE GENERATING MECHANISM

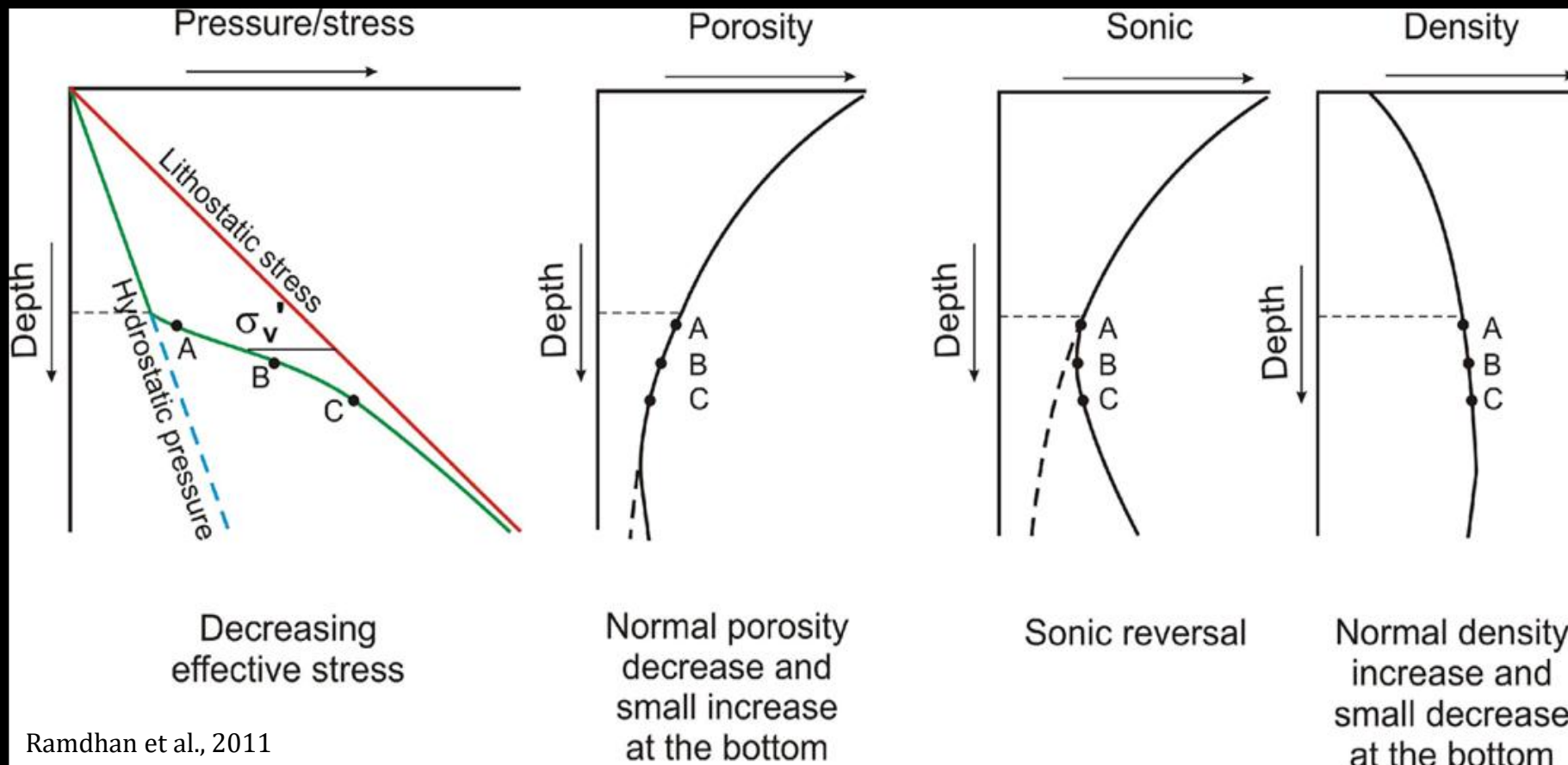


Loading/Disequilibrium Compaction

Ramdhan et al., 2011

This occurs when there is a failure in the compaction process due to the inability of fluid to drain from the shale pores when stress increases. This commonly results from rapid deposition rate that exceed the rate at which water can be expelled from sedimentary rocks.

OVERPRESSURE GENERATING MECHANISM



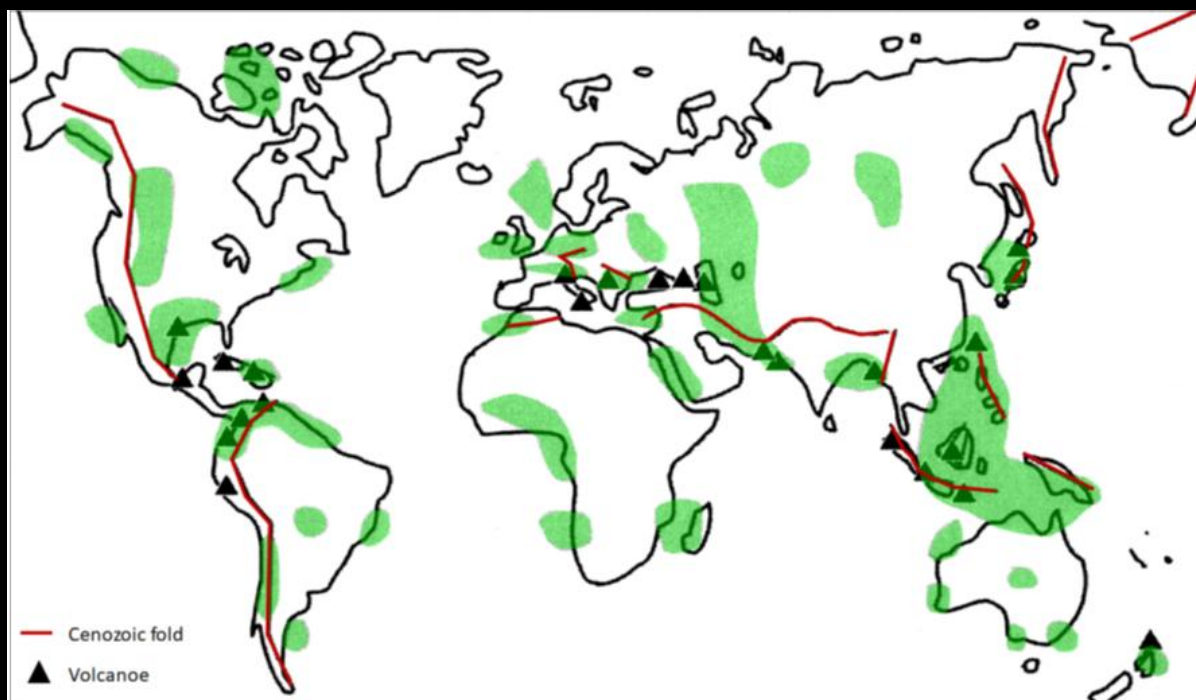
Unloading/Non-loading/Fluid Expansion

This results from a shift in pressure caused by the pressure transfer from the grain-to-fluid contact within the pore spaces such as from hydraulic head, hydrocarbon generation, lateral or vertical pressure transfer, clay mineral diagenesis, and an increase in fluid temperature.

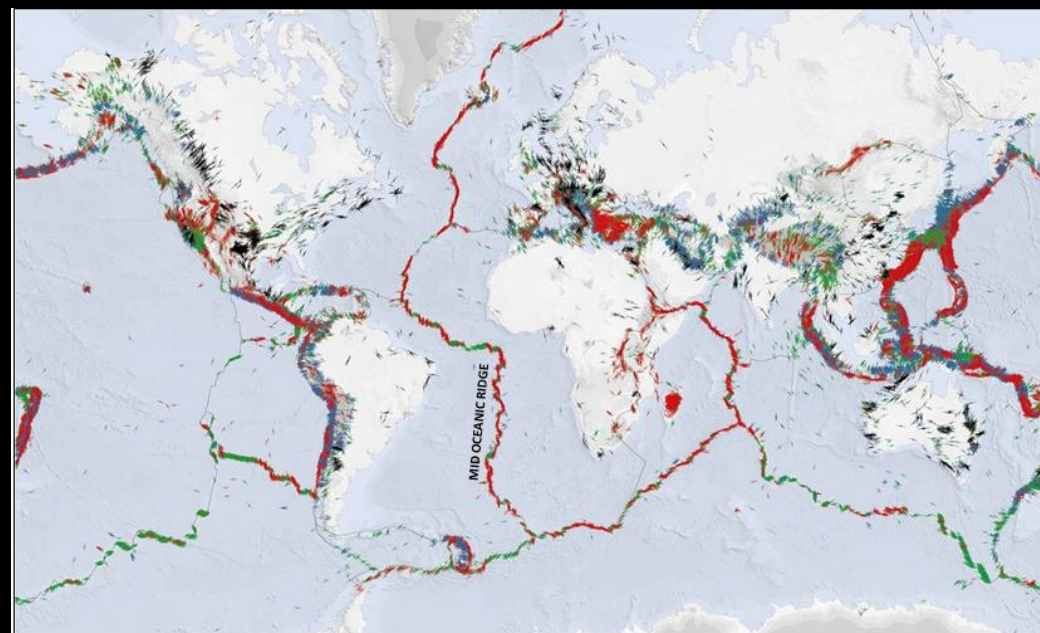
OVERPRESSURE GENERATING MECHANISM

Tectonic

Overpressure can be produced by compressional tectonics in both loading and non-loading scenarios. During the loading process, tectonic activity and sediment deposition occur simultaneously, potentially leading to under-compaction or disequilibrium compaction. However, when tectonic uplift is followed by erosion, the non-loading process takes place, changing pressure due to the transfer of contact pressure between grains into the pore fluid.

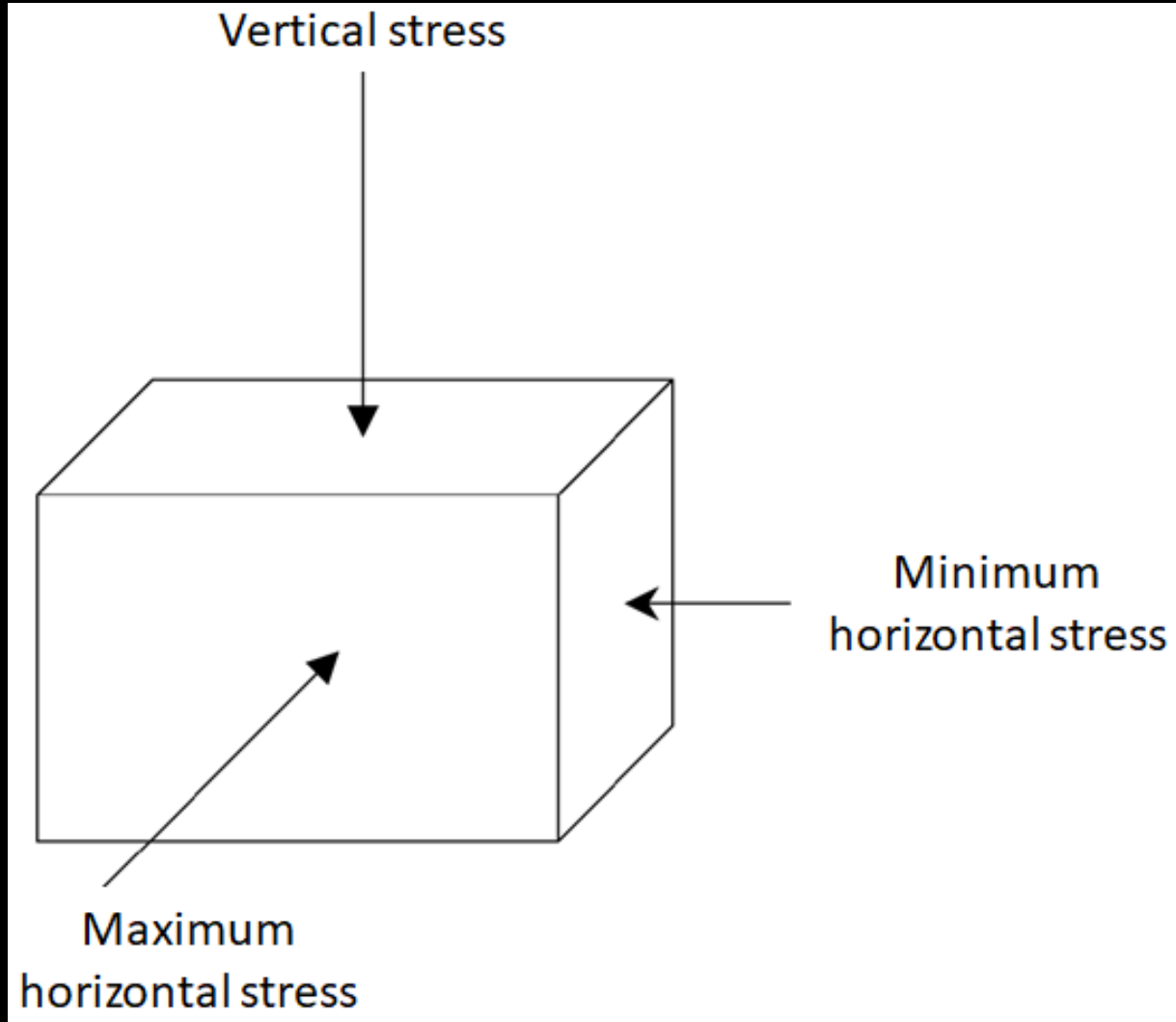


Schematic map of global overpressure occurrences (light green shading), areas of Cenozoic folding (dark red lines), and deep-seated mud volcanoes (triangles). Modified from Yasir and Addis, 2002 – AAPG memoir 76



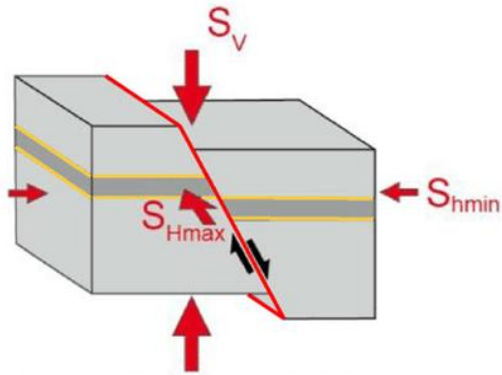
World Stress Map 2025. Lines show the orientation of maximum horizontal stress SH_{max} . The colours indicate the three classic stress regimes. Red indicates normal faulting stress regime in which the vertical stress SV is greater than SH_{max} and greater than the minimum horizontal stress SH_{min} ($SV > SH_{max} > SH_{min}$). Green are data points with strike-slip stress regime ($SH_{max} > SV > SH_{min}$) and blue with thrust faulting stress regime ($SH_{max} > SH_{min} > SV$). (Source: <https://doi.org/10.5880/WSM.2025.002>)

PRESSURE/STRESS CONCEPTS



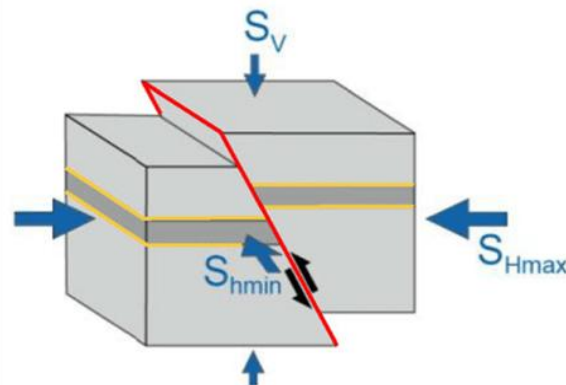
- In-situ stress refers to the natural state of stress present within the Earth's crust before any human activities, such as drilling, production, or fluid injection, alter the stress field.
- These stresses develop over geological timescales due to processes like tectonic plate movements, sedimentation, and gravitational loading.
- In-situ stress is typically described in terms of three principal components: the vertical stress (S_v), the maximum horizontal stress (S_{Hmax}), and the minimum horizontal stress (S_{Hmin}).

PRESSURE/STRESS CONCEPTS

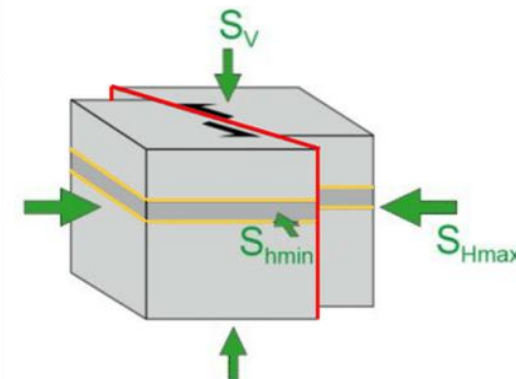


Normal fault regime
 $S_v > S_{Hmax} > S_{hmin}$
 $S_v = S_1$

A stress regime describes the relative magnitudes and orientations of the three principal in-situ stresses that act within the Earth's crust. It defines how the Earth's crust is being deformed, whether it's being stretched, compressed, or sheared, and it directly controls fracture orientation, fault behavior, and fluid flow in the subsurface.



Thrust fault regime
 $S_{Hmax} > S_{hmin} > S_v$
 $S_v = S_3$



Strike-slip fault regime
 $S_{Hmax} > S_v > S_{hmin}$
 $S_v = S_2$

PORE PRESSURE AND IN-SITU STRESS INTERACTION

Effective Stress Principle:

- ✓ The difference between total stress and pore pressure determines rock strength and deformation (Terzaghi's principle, 1936).

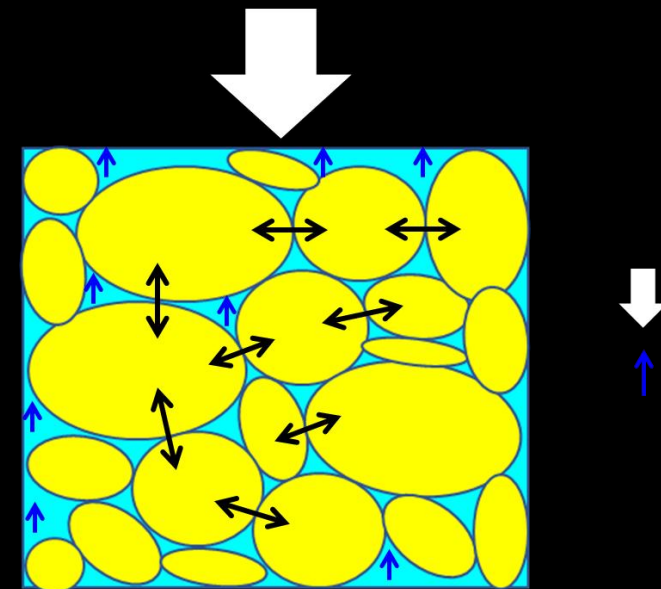
$$\sigma' = \sigma - Pp$$

where:

σ' = effective stress

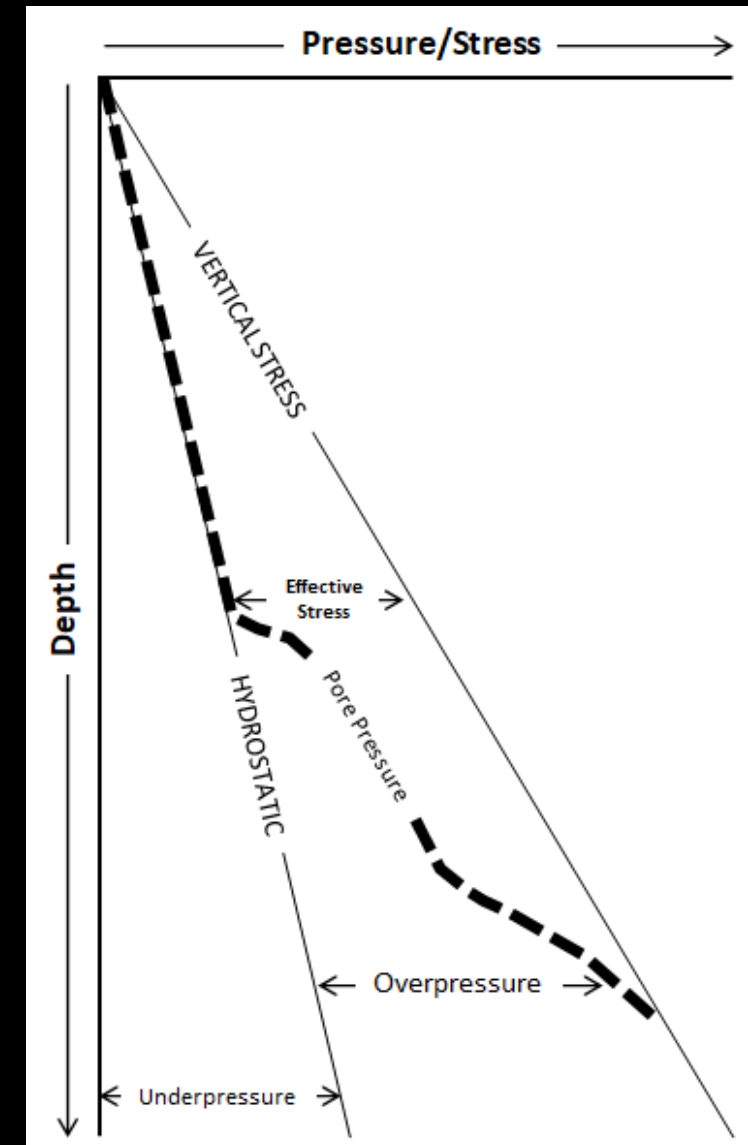
σ = total stress (vertical or overburden)

Pp = pore pressure.



PORE PRESSURE AND IN-SITU STRESS INTERACTION

- **Normal Pressure Environments:**
 - Pore pressure equals hydrostatic pressure.
 - Effective stress increases linearly with depth.
- **Overpressure Environments:**
 - Pore pressure exceeds hydrostatic pressure.
 - Constant or Reduction in effective stress.
 - Common in rapidly buried sediments or tectonically active regions.
- **Implications for Drilling:**
 - Drilling into overpressured zones can lead to catastrophic well failures if not anticipated.



DIRECT STRESS MEASUREMENT: MINIMUM HORIZONTAL STRESS

Typical XLOT plot

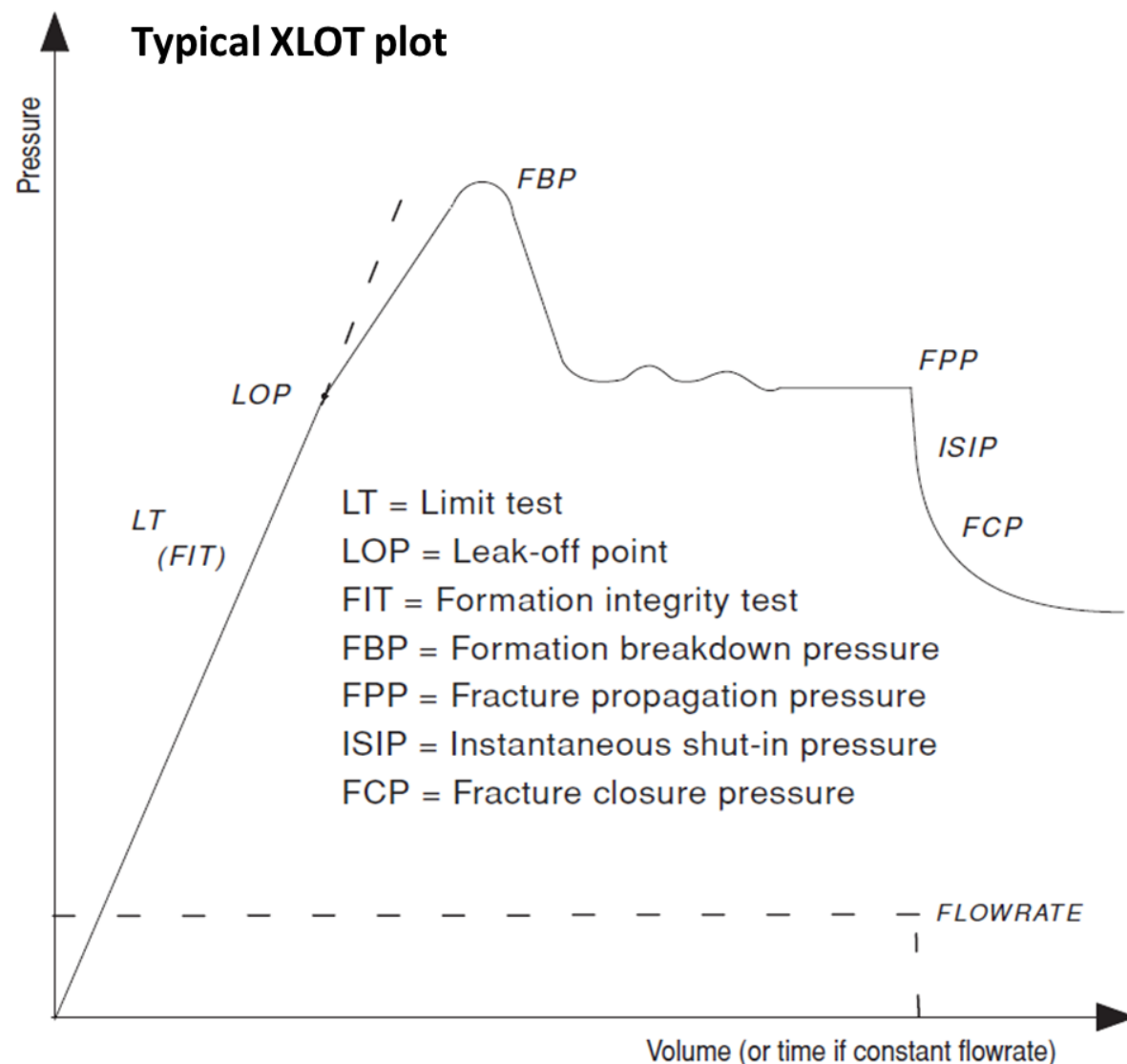


Illustration of Leak-Off Test (LOT) and Formation Integrity Test (FIT) at a well (Zoback, 2007).

- Leak-Off Point (LOP) is achieved when the rock formation in the well hole begins to break due to pressure on the well hole.
- When formation breakdown pressure (FBP) conditions are reached, the pressure on the well hole will decrease and then stabilize as a result of the flow of drilling fluid into the formation.
- A stable pressure with a pump that is still active is called Fracture Propagation Pressure (FPP).
- When the pump is turned off, the pressure on the well hole will begin to decrease called Initial Shut-In Pressure (ISIP).
- The pressure on the well hole will continue to decrease until the fractures close so called Fracture Closure Pressure (FCP).
- S_{hmin} magnitude is estimated based on the amount of pressure at the time FPP is reached which is not much different from LOP.

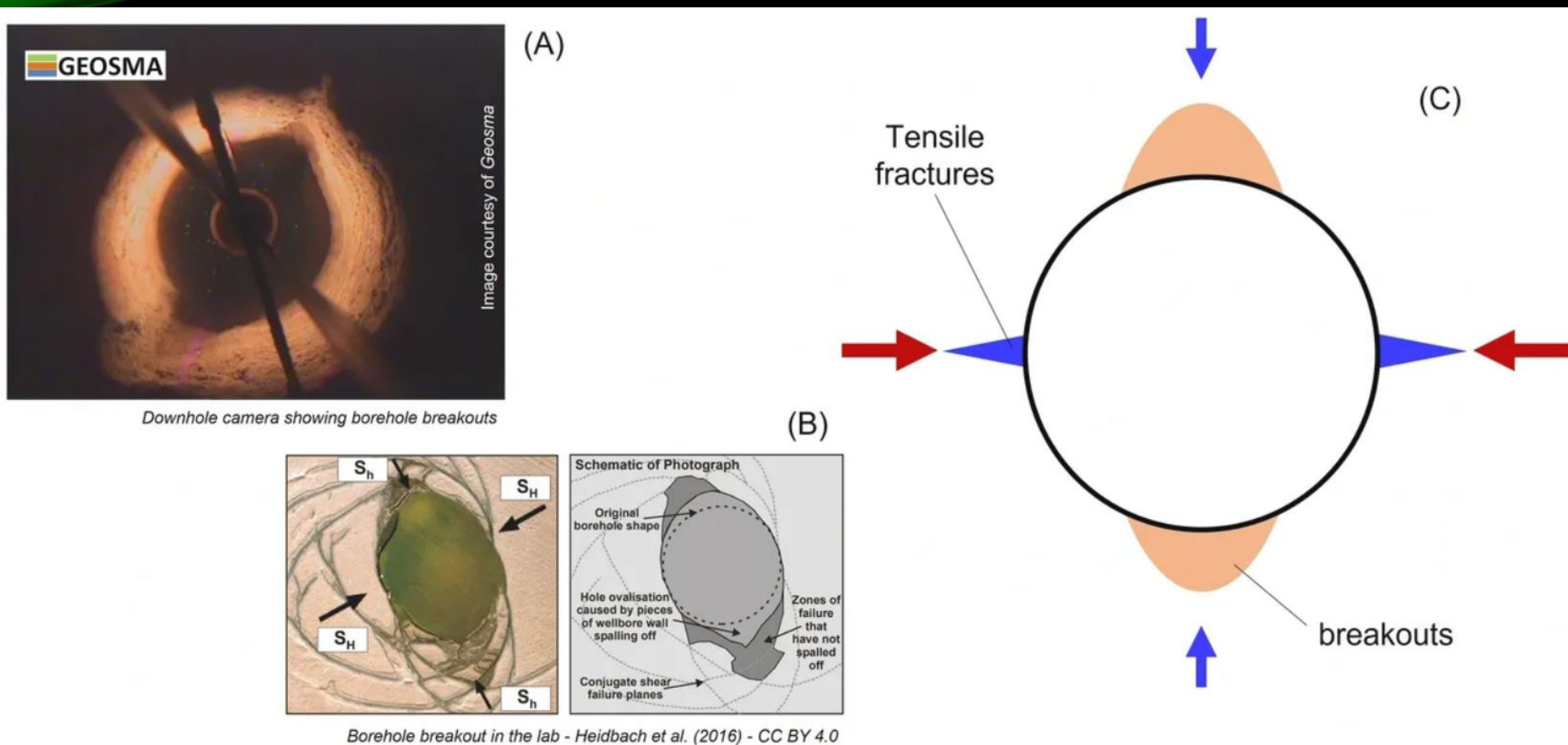
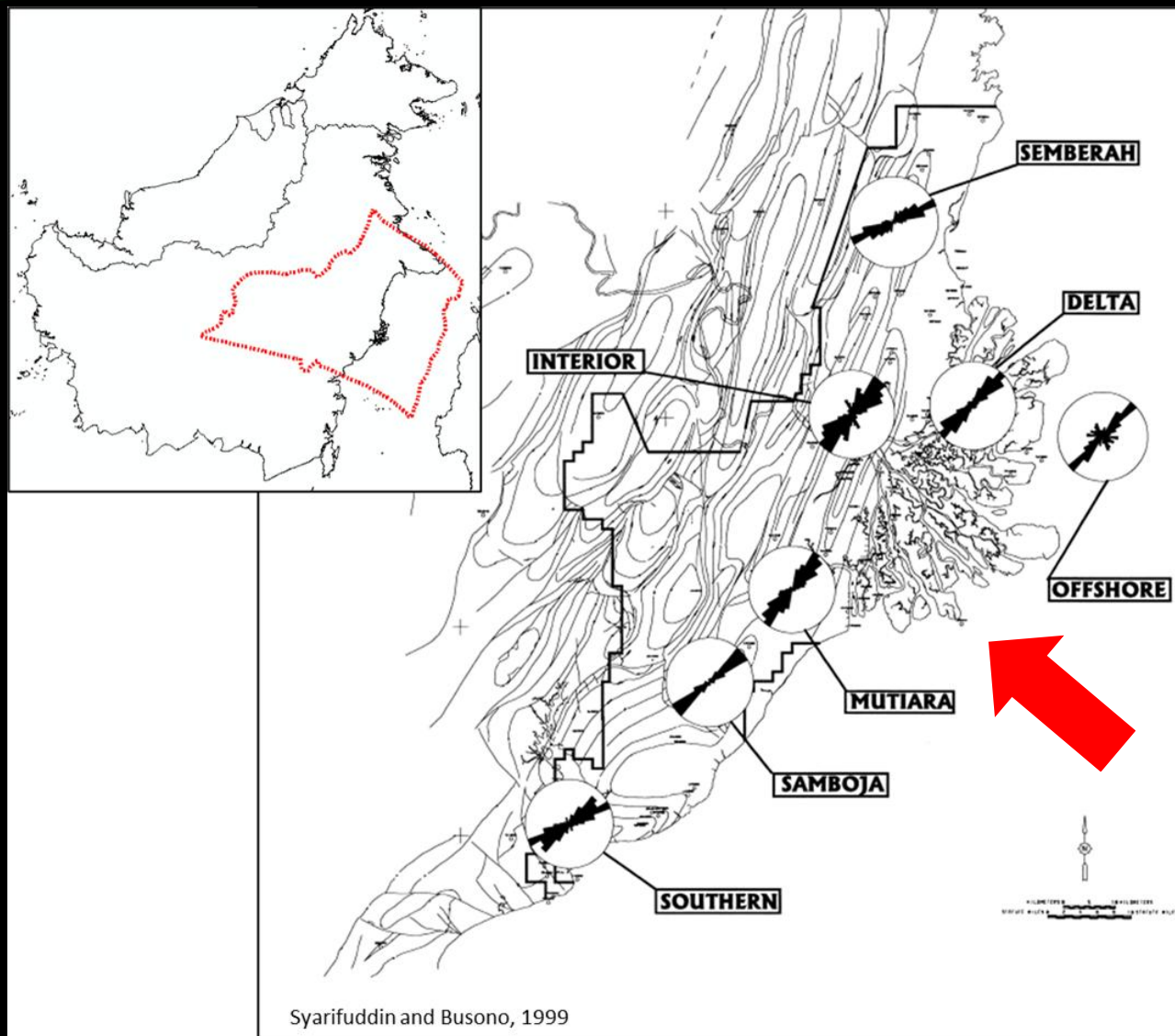
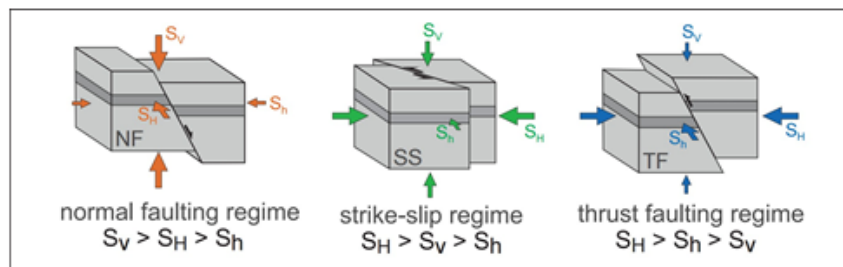


Figure 3. (A) Example of wellbore breakout seen by a downhole camera (courtesy of GEOSMA); (B) development of borehole breakouts in a laboratory experiment (Heidbach et al, 2018); (C) Scheme showing occurrence of tensile fractures and borehole breakouts with respect to S_{Hmax} orientation (red arrow) in a vertical wellbore.

IN-SITU STRESS CHARACTERISTICS IN KUTAI BASIN

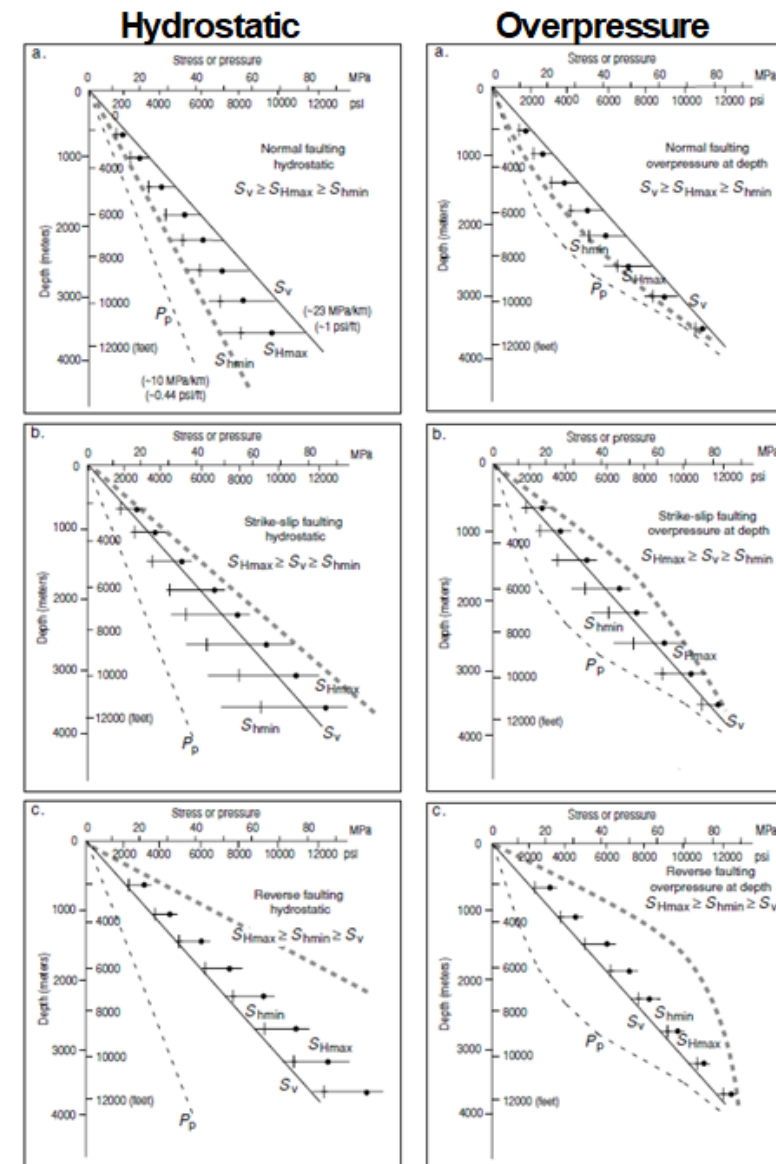


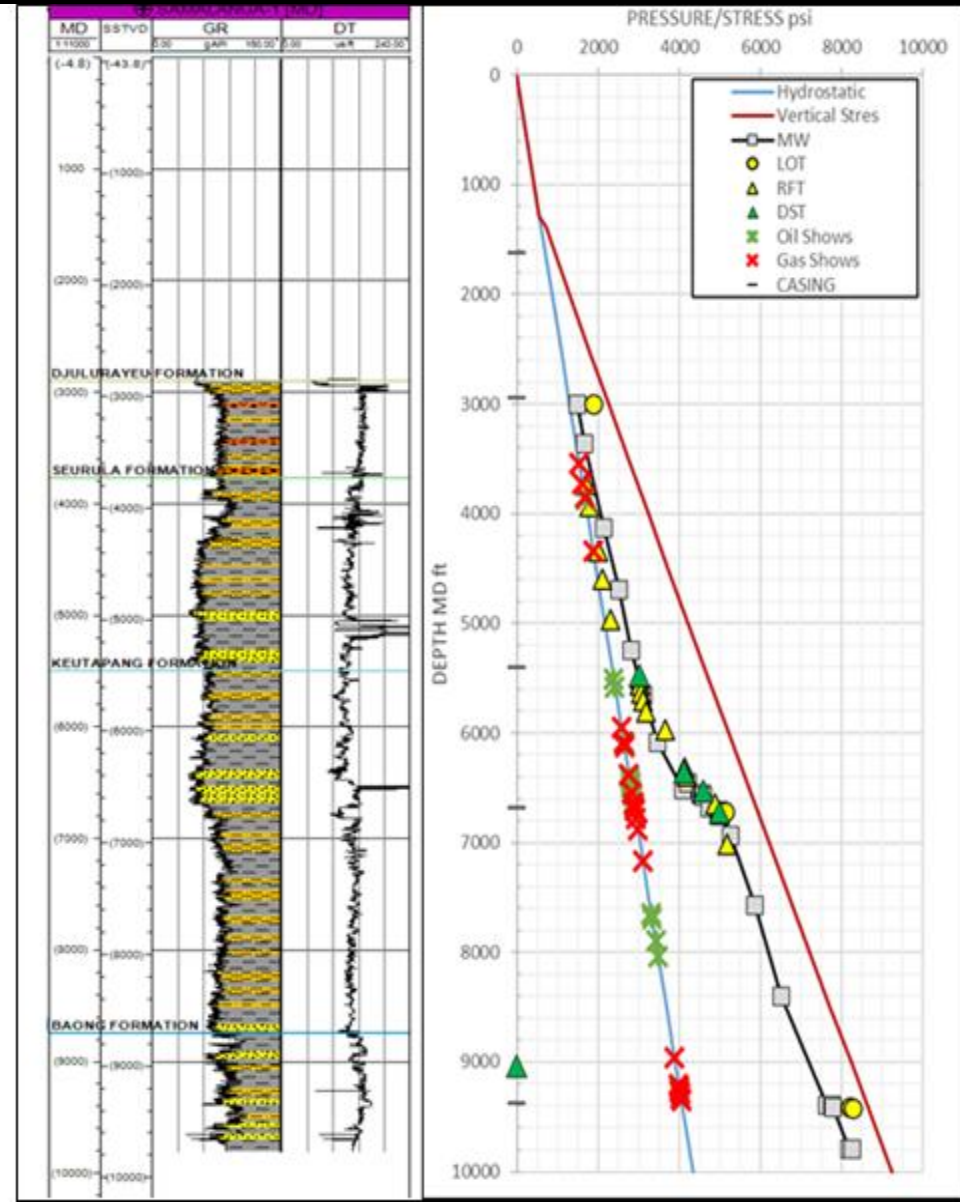
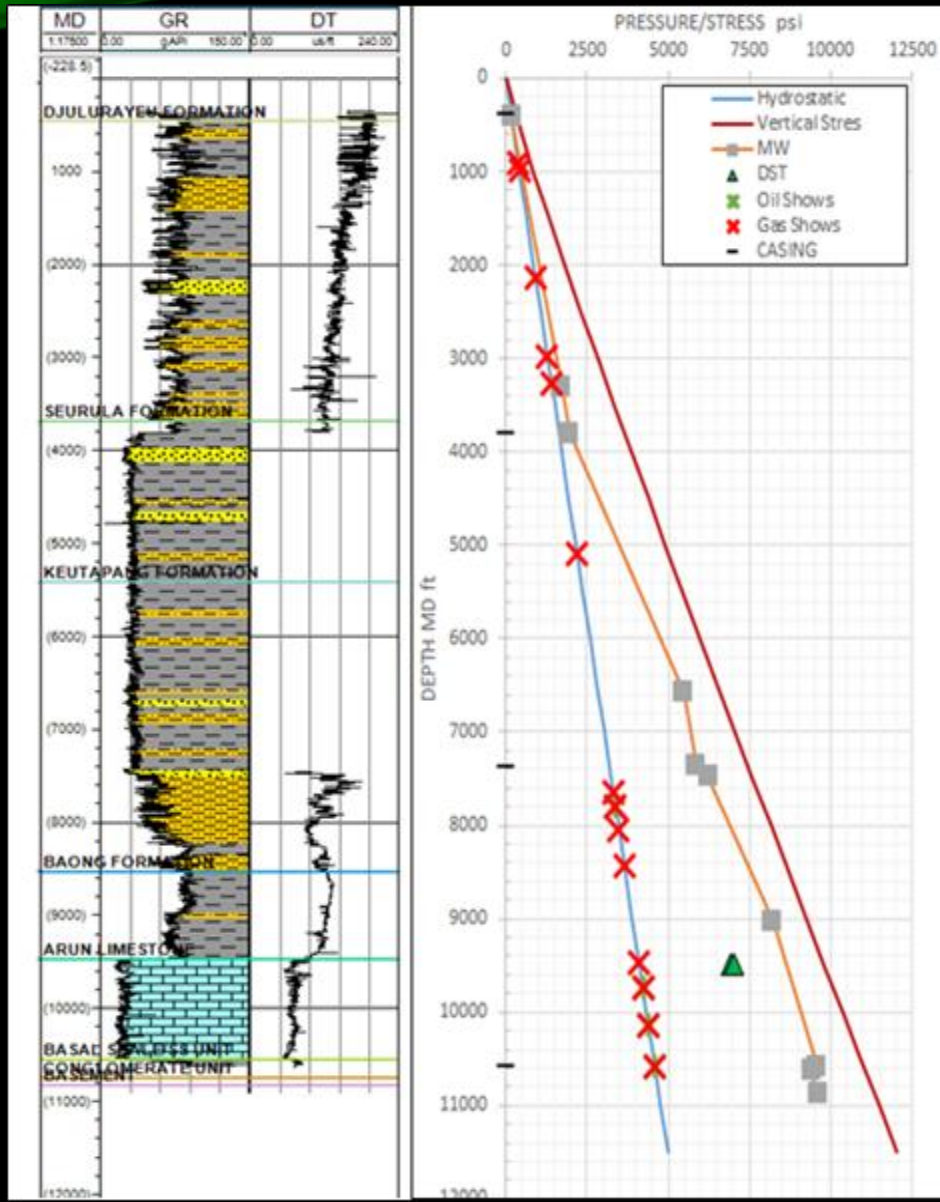
- Map showing the mean azimuth for borehole breakout orientations from 134 wells.
- The breakout azimuths show a relatively consistent NE±SW alignment which indicates that **the maximum regional horizontal stress direction is in NW±SE direction**
- The breakout azimuths are perpendicular to the axis of basin shortening. These relationships indicate strongly that a compressive stress system is still active and is responsible for the borehole breakouts.



Regime	Stress		
	S_1	S_2	S_3
Normal	S_v	S_{Hmax}	S_{hmin}
Strike-slip	S_{Hmax}	S_v	S_{hmin}
Reverse	S_{Hmax}	S_{hmin}	S_v

- Variation of stress magnitudes with depth in normal, strike-slip and reverse faulting stress regimes for hydrostatic and overpressure conditions.
- The difference between principal stresses increases with depth due to the increase of the frictional crustal strength of the crust with depth, but decreases as severe overpressure develops due to the decrease of frictional strength with elevated pore pressure.





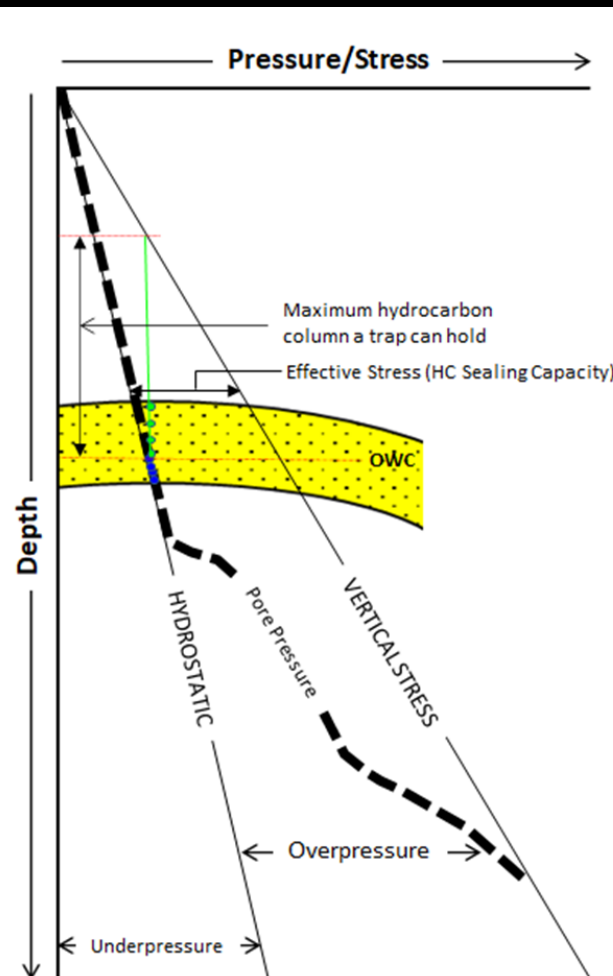
EXPLORATION CASE

Sealing Capacity

Sealing capacity refers to the ability of a geological formation, especially a cap rock, to prevent the migration of hydrocarbons or other fluids. The sealing capacity is typically governed by factors like:

- The permeability of the rock (whether it's low enough to prevent fluid flow),
- The integrity of the rock (no fractures or faults that could allow fluid escape),
- Capillary barriers that can prevent the upward migration of fluids, such as oil or gas.

The sealing capacity is a key factor in the formation of traps that contain hydrocarbons and prevent their migration to the surface or to other formations.



Vertical Effective Stress:

$$\sigma_v' = \sigma_v - PP$$

Maximum Hydrocarbon Column Height:

$$h_{hc} = \sigma_v' / (\rho_w - \rho_{hc}) \cdot g$$

- σ_v : Vertical stress (MPa)
- PP : Pore pressure (Mpa)
- σ_v' : Vertical Effective Stress (MPa)
- h_{hc} : maximum hydrocarbon column height (m)
- ρ_w : Water density (e.g., 1 g/cm³ or 1000 kg/m³)
- ρ_{gas} : gas density (e.g., 0.2 g/cm³ or 200 kg/m³)
- g : gravity (9.81 m/s²)

Fluid Densities (Typical Values)

Brine Water:

$$\rho_w = 1.03 \text{ g/cm}^3 \text{ (or 8.58 ppg)}$$

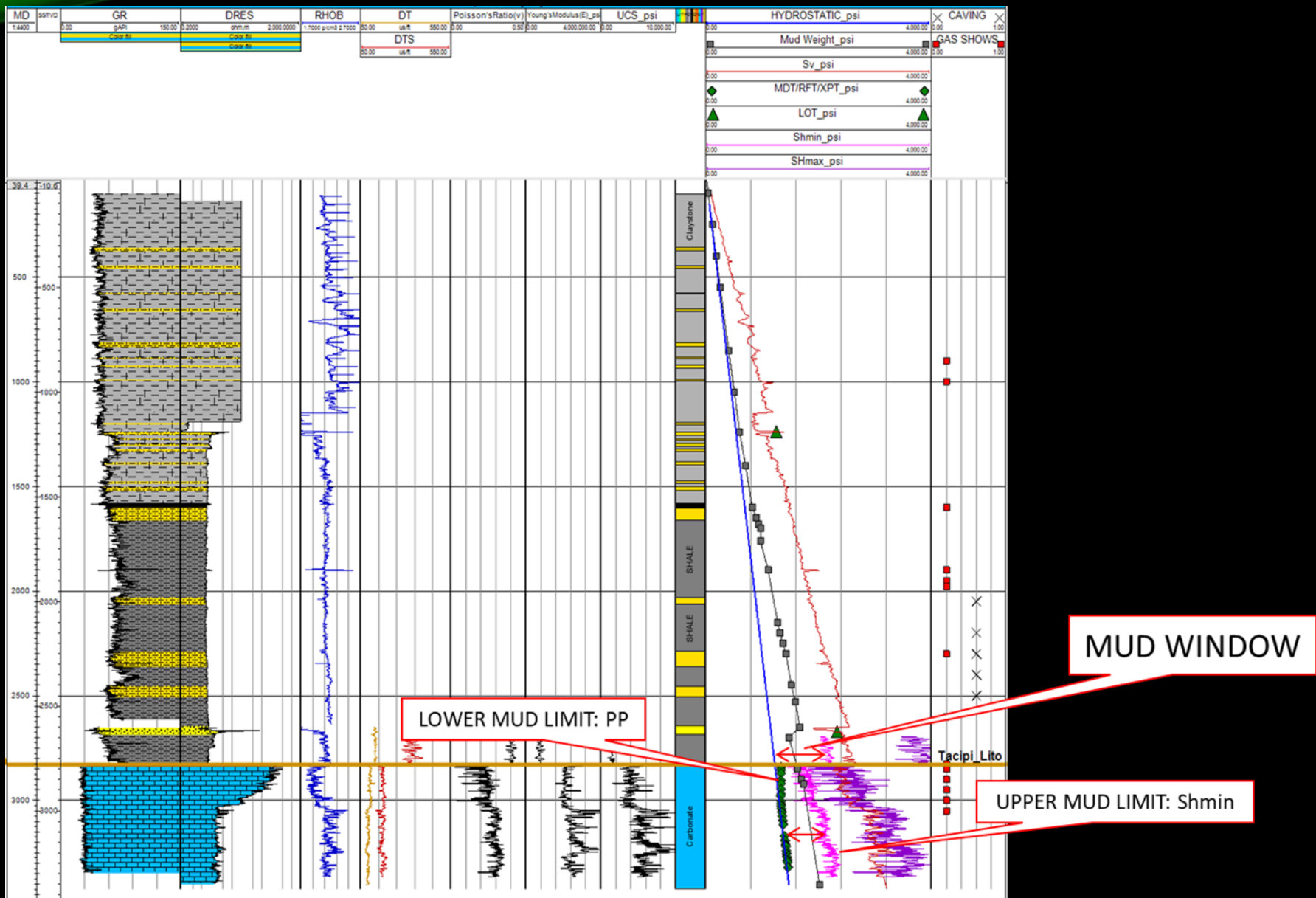
Hydrocarbon (Gas or Oil):

$$\rho_{gas} \approx 0.2 \text{ g/cm}^3 \text{ (or 1.67 ppg)}$$

$$\rho_{oil} \approx 0.8 \text{ g/cm}^3 \text{ (or 6.65 ppg)}$$



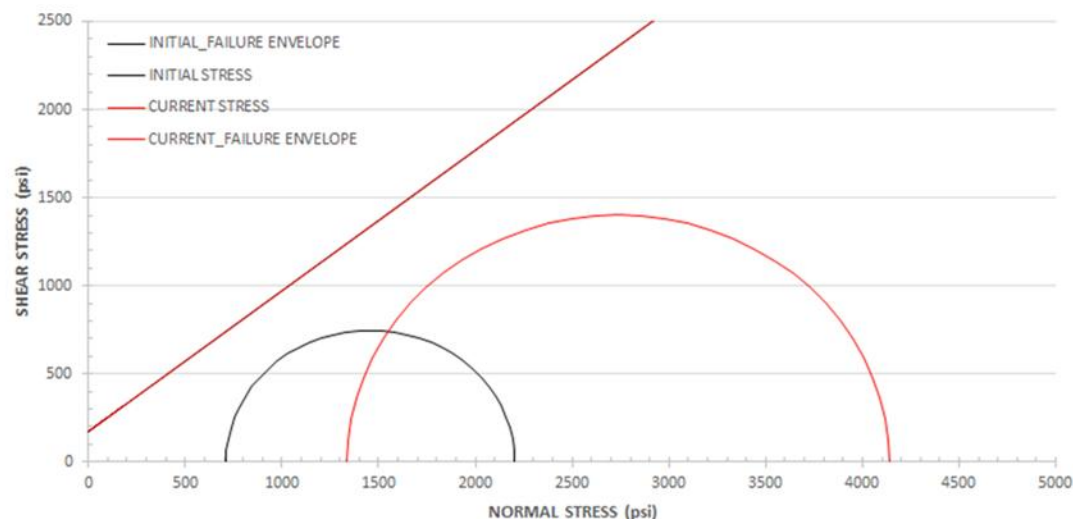
SAFE MUD WINDOW



CASE 1

NO CHANGE IN COHESION AND FRICTION ANGLE DUE TO PRODUCTION

FAILURE ANALYSIS

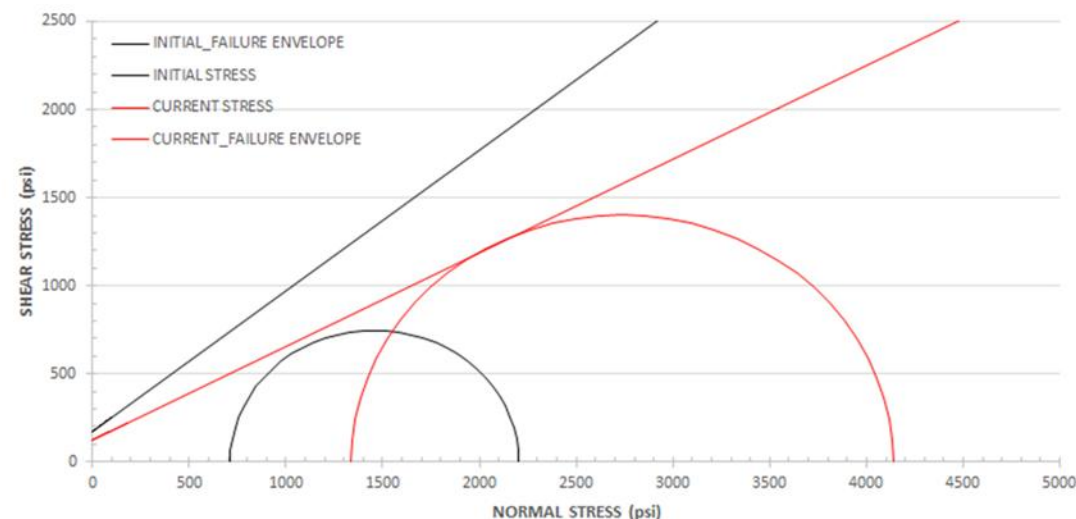


INPUT INITIAL		INPUT CURRENT	
Parameter		Parameter	
Sv_psi	2169	Sv_psi	2169
Shmax_psi	3582	SHmax_psi	4501
Shmin_psi	2089	Shmin_psi	1697
Pp_psi	1378	Pp_psi	362
Cohesion c_psi	174	Cohesion c_psi	174
Friction Angle deg.	38.6	Friction Angle	38.6
MW_psi	1400	MW_psi	1400
PR	0.3	PR	0.3
Biot Coeff.	0.9	Biot Coeff.	0.9

CASE 2

DECREASE IN COHESION AND FRICTION ANGLE DUE TO PRODUCTION

FAILURE ANALYSIS



INPUT INITIAL		INPUT	
Parameter		Parameter	
Sv_psi	2169	Sv_psi	2169
Shmax_psi	3582	SHmax_psi	4501
Shmin_psi	2089	Shmin_psi	1697
Pp_psi	1378	Pp_psi	362
Cohesion c_psi	174	Cohesion c_psi	125
Friction Angle deg.	38.6	Friction Angle	28
MW_psi	1400	MW_psi	1400
PR	0.3	PR	0.3
Biot Coeff.	0.9	Biot Coeff.	0.9



PRE-SPUD MEETING

Eastern Indonesia exploration well (2006) Final Drilling Report:

6. Geological Uncertainty Notes

The original Geological Prognosis

Please refer to the Drilling program, however, it is reproduced below for information.

Notes;

Tectonic stresses and overpressure were not prognosed in [REDACTED] The Tectonic stresses/overpressure caused the vast amount of the lost time in this well, requiring mud weights and casing seat strengths in excess of offset well data.



RISKING



RISKING – GEOLOGICAL CHANCE FACTOR (POSG)

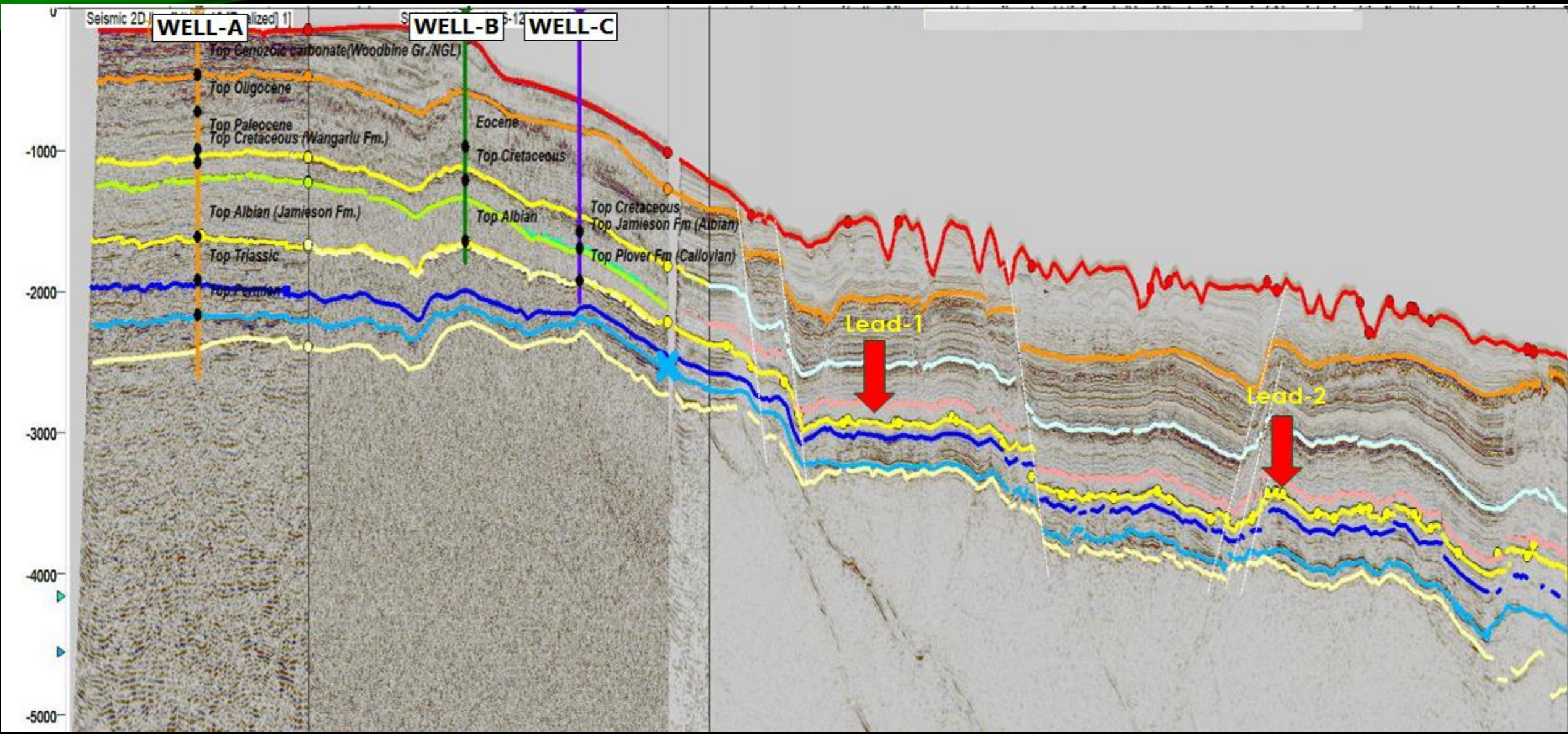
Geological Chance Factor (Posg) =

Psource x Pgeneration x Pmigration x Preservoir x Ptrap x Pseal



CASE STUDY

EXPLORATION CASE



VOLUMETRIC CALCULATION

OOIP (Original Oil in Place) Formula

The common formula for calculating OOIP in barrels (stb) is:

$$OOIP = \frac{7758 \times A \times h \times \phi \times (1 - S_w)}{B_{oi}}$$

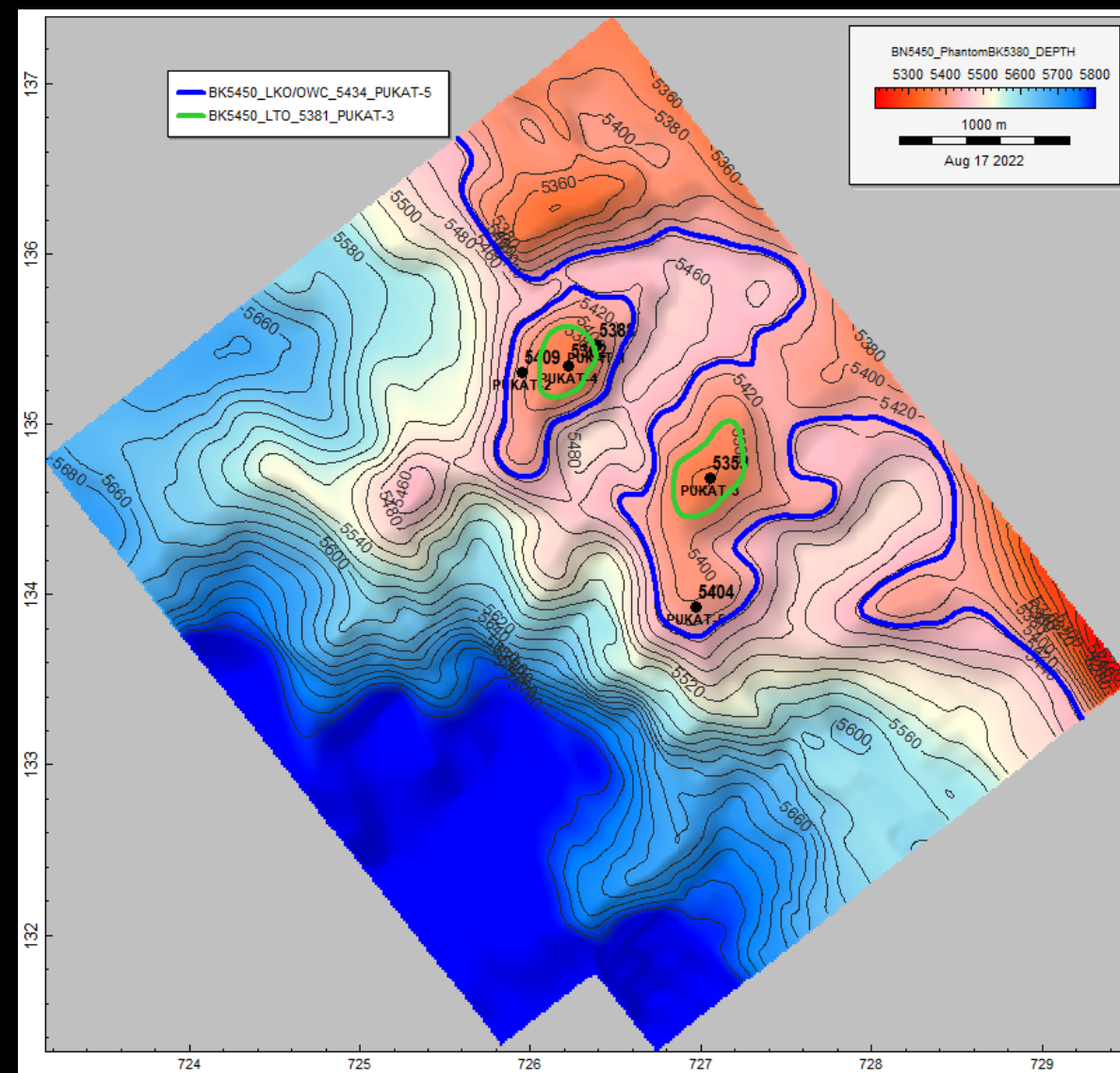
- **7758**: Conversion factor from acre-ft to barrels (bbl).
- **A**: Reservoir Area (acres).
- **h**: Net Pay Thickness (feet).
- **ϕ** : Porosity (decimal).
- **S_w** : Water Saturation (decimal).
- **B_{oi}** : Oil Formation Volume Factor (rb/stb). wiki.aapg.org

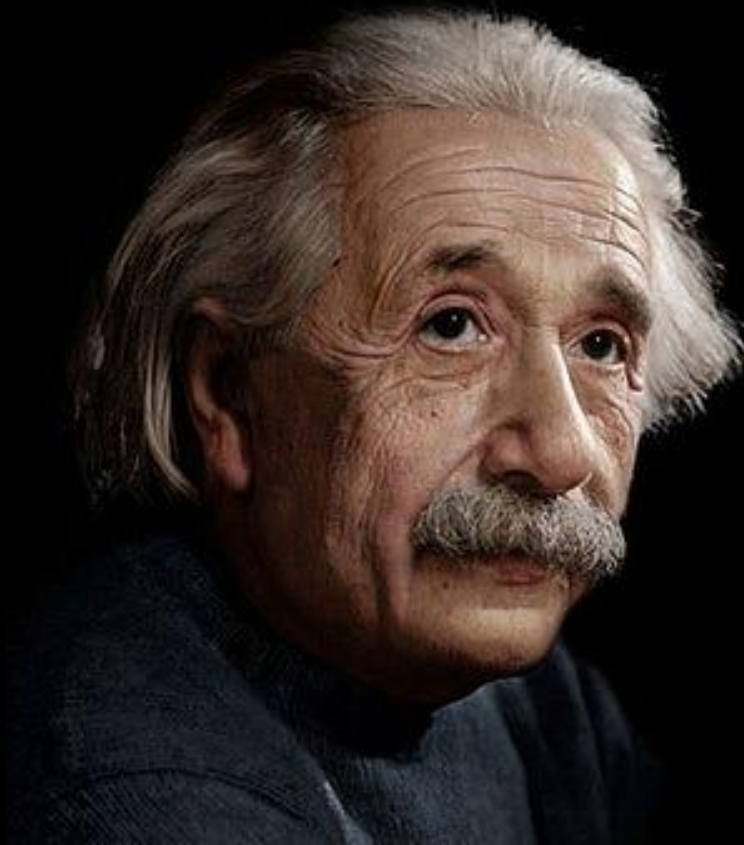
OGIP (Original Gas in Place) Formula

The common formula for calculating OGIP in standard cubic feet (scf) is:

$$OGIP = \frac{43560 \times A \times h \times \phi \times (1 - S_w)}{B_{gi}}$$

- **43560**: Conversion factor from acre-ft to ft³.
- **B_{gi}** : Gas Formation Volume Factor (ft³/scf).





Imagination is more
important than knowledge.
Knowledge is limited.
Imagination encircles
the world.

Albert Einstein / @InspiringThinkn